

# COCO: A Cohesiveness-aware Learning Framework for Community Search over Temporal Graphs

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## Abstract

In this paper, we explore time-interval-based community search over temporal graphs, which aims to identify a connected cohesive subgraph containing the query vertices within a specific time interval. Existing work suffers from two main limitations: (1) *information loss*, as it ignores the interaction frequency between vertices, a crucial factor for identifying high-quality communities. (2) *efficiency bottleneck*, as it either incurs high search latency from expensive online computation or faces a combinatorial explosion in indexing time and storage when precomputing coreness values across all possible time intervals. To address these issues, we propose COCO, a cohesiveness-aware learning framework for time-interval-based community search over temporal graphs. COCO introduces a new coupling mechanism between interval-specific coreness computation and query-specific temporal representation learning: a *hierarchical model index* predicts coreness for arbitrary time intervals via a hierarchical model tree, and an *adaptive temporal GNN* uses the predicted coreness as a structural prior while capturing interaction frequencies through query-specific fine-tuning. Experiments on eight real-world datasets show that COCO outperforms state-of-the-art methods in community quality and is at least 10× faster than the baselines, demonstrating its effectiveness and efficiency.

## 1 Introduction

Community search is an important tool in graph analysis, aiming to identify cohesive subgraphs that contain given query vertices [6, 11, 26, 40]. Prior research has extensively investigated community search on various graph types, including directed graphs [3, 7, 22], bipartite graphs [37, 38], attributed graphs [23, 41], and multilayer graphs [2, 24, 33].

In real-world applications, interactions between entities are dynamic and evolve over time. For example, in social networks, user activities such as comments, likes, and shares are timestamped, which reflects the temporal nature of relationships. This temporal information provides an important view for understanding user behavior patterns and community structure. As a result, community search over temporal graphs has attracted considerable research attention [18–21, 43, 46]. For example, QTCS [21] is a state-of-the-art temporal community search method that uses a query-centered  $k$ -core variant, the  $\beta$ -temporal proximity core. In this core, each vertex’s query-biased temporal degree (the sum of its neighbors’ temporal-proximity scores to the query vertex) is at least  $\beta$ . The

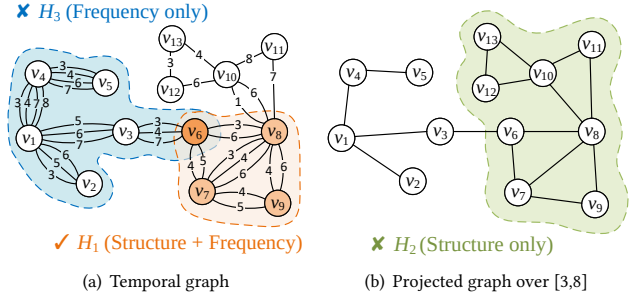


Figure 1: A motivating example

returned community retains a core structure while remaining temporally close to the query vertex. For community search over a specified time interval, [43] proposes the temporal  $k$ -core method. Given a temporal graph, a query vertex, a time interval, and an integer  $k$ , it returns a connected  $k$ -core containing the query vertex in a projected graph. This projected graph is constructed by retaining only the edges within the specified time interval, where multiple edges between the same pair of vertices are merged into one.

Figure 1(a) shows a temporal graph in which each edge is timestamped to indicate when the corresponding interaction occurred. Given the query vertex  $v_6$ , the time interval [3, 8], and  $k = 2$ , the method in [43] identifies the temporal 2-core  $H_2 = \{v_6, \dots, v_{13}\}$  in the corresponding projected graph, as shown in Figure 1(b). Here, QTCS is applied to the temporal graph restricted to [3, 8]. For the same query vertex  $v_6$ , it also returns  $H_2$ .

**Limitations.** When performing time-interval-based community search, these methods exhibit two major limitations.

**L1. Information Loss.** In Figure 1, the temporal  $k$ -core method [43] and QTCS [21] both return the vertex set  $H_2 = \{v_6, \dots, v_{13}\}$  for query vertex  $v_6$ . Within [3, 8], every vertex pair involving  $v_{10}$ ,  $v_{11}$ ,  $v_{12}$ , or  $v_{13}$  interacts only once. Both methods emphasize structural cohesiveness, while QTCS additionally considers temporal proximity to the query vertex. Nevertheless, both include these less active vertices. Interaction frequency (the number of distinct temporal edges between two vertices) distinguishes single interactions from repeated ones and measures connection strength. However, using only interaction frequency can produce a community with a weak structure. Every edge in  $H_3 = \{v_1, \dots, v_6\}$  occurs at least three times, but  $H_3$  is a tree. The query vertex  $v_6$  is connected to the remaining vertices only through  $(v_3, v_6)$ . In contrast, the community  $H_1 = \{v_6, v_7, v_8, v_9\}$  is better than  $H_2$  and  $H_3$  because it considers both structural cohesiveness and interaction frequency.

**L2. Efficiency Bottleneck.** Both the method in [43] and QTCS [21] have serious efficiency limitations. Performing online core decomposition for each query interval is computationally expensive and leads to high search latency on large-scale graphs. For example, on graphs with millions of vertices, online  $k$ -core decomposition may take around 500 seconds, while QTCS, which processes each

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query from scratch, may take several hundred seconds (see Section 6.2 for details). Although indexing can accelerate temporal  $k$ -core queries, it requires precomputing and storing results for all possible  $k$  values and time intervals. Since a temporal graph with  $t_{\max}$  timestamps contains  $O(t_{\max}^2)$  continuous intervals, this approach incurs prohibitive preprocessing time and memory overhead, making it impractical for large-scale temporal graphs. In particular, the index for a graph with two million vertices can exceed 1 GB, and for even larger graphs, it cannot be constructed within 24 hours (see Section 6.2 for details).

To address these limitations, we revisit and formalize the *Time-Interval-based Community Search (TICS)* problem. Specifically, given a temporal graph  $\mathcal{G}$ , a time interval  $[t_s, t_e]$ , and a set of query vertices  $V_Q$ , the goal of TICS is to identify a community containing  $V_Q$  within  $[t_s, t_e]$ .

**Challenges.** The TICS problem presents two main challenges.

**C1. How to identify high-quality communities?** High-quality temporal communities should exhibit both cohesive structural topology and frequent temporal interactions. However, integrating these two heterogeneous signals is intrinsically difficult. Existing structure-based cohesive subgraphs cannot effectively measure interaction frequency, while purely frequency-based metrics often overlook the overall specific community structures.

**C2. How to address TICS efficiently?** The time interval of each TICS query may differ, and therefore the graph topology associated with each query can also be different. Processing every TICS query from scratch is computationally expensive, especially on large-scale temporal graphs. Although preprocessing or indexing may help, such methods must handle the large number of possible graph structures across all time intervals, which further increases the computational cost.

**Our Solutions.** To overcome these challenges, we propose **COCO**, a novel **CO**hesiveness-aware learning framework for **CO**munity search over temporal graphs. COCO introduces a new coupling mechanism between interval-specific coreness computation and query-specific temporal representation learning. This mechanism is realized through a directed information flow in which coreness serves as an explicit structural prior during model training and query-specific adaptation, thereby tightly integrating the classical  $k$ -core model with graph neural networks (GNNs). COCO offers two key advantages: (1) it identifies high-quality communities with both strong structural cohesiveness and frequent interactions; and (2) it efficiently processes queries over arbitrary time intervals without recomputing  $k$ -core or retraining the entire model for each query.

Specifically, COCO incorporates two closely connected components. (1) The **Hierarchical Model Index (HM-Index)** is a tree-structured index for predicting vertex coreness over arbitrary time intervals (**for C2**). To the best of our knowledge, HM-Index is the first approach that avoids exhaustively computing coreness values over all  $O(t_{\max}^2)$  possible time intervals. It recursively partitions the entire temporal span into nested intervals, each represented by a tree node containing a lightweight prediction model. For a query interval, HM-Index traverses the tree from the root and predicts coreness through coarse-to-fine refinement. Each child model refines its parent's prediction using more local interval information, improving coreness prediction accuracy. (2) The **Adaptive**

**Temporal GNN (AT-GNN)** is a query-adaptive GNN for capturing structural cohesiveness and interaction frequency (**for C1**). It combines the coreness predicted by HM-Index with temporal interaction features, tightly coupling the discrete structural information of the  $k$ -core model with continuous temporal representation learning. To handle different query structures, AT-GNN inserts lightweight query-aware adapters into each encoder layer. For each query, only the adapters are fine-tuned, enabling query-specific adjustment without retraining the full model (**for C2**).

COCO operates through a three-phase pipeline. First, in the pre-training phase, COCO trains coreness prediction models on different time intervals and then organizes these models into an HM-Index, and trains AT-GNN in an unsupervised manner, enabling it to learn common community patterns across time intervals. In the fine-tuning phase, COCO predicts the coreness of query vertices and their neighbors by traversing the HM-Index, and uses these coreness values along with the query context to adjust the adapter module parameters of AT-GNN. In the searching phase, COCO uses vertex embeddings from AT-GNN to iteratively select vertices most similar to the query vertices, forming the final community.

**Contributions.** Overall, our main contributions are as follows.

- We propose COCO, a cohesiveness-aware learning framework for TICS. COCO introduces a coupling mechanism between interval-specific coreness computation and GNN-based temporal representation learning to efficiently identify high-quality temporal communities.
- We design the *Hierarchical Model Index* (HM-Index), a learning-based index that arranges lightweight models in a hierarchical tree. Its coarse-to-fine refinement predicts coreness over arbitrary time intervals without computing coreness for all  $O(t_{\max}^2)$  intervals, reducing preprocessing time and storage costs.
- We develop the *Adaptive Temporal GNN* (AT-GNN) to jointly capture graph structure and interaction frequency. Its query-aware adapters enable fast adaptation to different query structures without full model retraining.
- We conduct extensive experimental studies on eight real-world graphs. The results validate the superior effectiveness and efficiency of our proposed COCO framework.

## 2 Related Work

We review the related work from two aspects, including classic community search over temporal graphs, and learning-based methods for community search. Table 1 compares representative temporal community search methods. COCO is the only method that jointly models cohesive subgraph structure and pairwise interaction frequency while natively supporting TICS.

**Classic community search over temporal graphs.** Based on temporal graph characteristics, various community search methods have been studied. Most active community search [18] uses a  $(k, t)$ -active core, while significant engagement community search [46] requires a connected  $k$ -core and maximizes the engagement of the query vertex. Persistent community search [19, 20] operates on snapshot sequences and seeks  $k$ -core communities that remain cohesive over time. QTCS [21] is a state-of-the-art method for query-centered temporal community search. It derives Time-Constrained Personalized PageRank (TPPR) scores from chronological walks

**Table 1: Comparison of temporal community search methods**

| Method                     | Underlying Model | Native TICS Support | Cohesiveness Modeling | Pairwise Interaction Frequency |
|----------------------------|------------------|---------------------|-----------------------|--------------------------------|
| Temporal $k$ -core [43]    | $k$ -core        | ✓                   | ✓                     | ✗                              |
| $(k, t)$ -active core [18] | $k$ -core        | ✗                   | ✓                     | ✗                              |
| SECS [46]                  | $k$ -core        | ✗                   | ✓                     | ✗                              |
| Persistent core [19, 20]   | $k$ -core        | ✗                   | ✓                     | ✗                              |
| QTCS [21]                  | $k$ -core + TPPR | ✗                   | ✓                     | ✗                              |
| CS-TGN [10]                | GNN              | ✗                   | ✗                     | ✗                              |
| UTCS [45]                  | node2vec         | ✗                   | ✗                     | ✗                              |
| PK-GCN [13]                | GNN              | ✗                   | ✓                     | ✗                              |
| COCO (our)                 | $k$ -core + GNN  | ✓                   | ✓                     | ✓                              |

and defines the query-biased temporal degree of a vertex as the sum of its neighbors' TPPR scores. QTCS then maximizes the minimum query-biased temporal degree in a connected  $\beta$ -temporal proximity core. This structural condition gives more weight to neighbors temporally close to the query vertex. For a specified time interval, temporal  $k$ -core component search [43] requires every vertex to have a projected degree of at least  $k$ . These methods do not directly measure the interaction frequency between vertex pairs. In contrast, COCO searches over a specified interval while jointly considering structural cohesiveness and interaction frequency within the returned community. Moreover, several temporal cohesive subgraph models are proposed, include span core [8, 9], periodic clique [28], periodic core [30], temporal  $k$ -core [42, 44, 47], and bursting core [29]. These methods are not specifically designed for query-vertex-centered community search.

**Learning-based methods for community search.** Learning-based community search has been studied in supervised settings using ground-truth communities [5, 12, 36], semi-supervised settings using partial labels [17, 32], and unsupervised settings using contrastive objectives or pseudo-labels [35, 39]. Most of these methods target static graphs and do not directly address TICS over arbitrary query intervals.

CS-TGN constructs query-dependent representations from local and global snapshot structures [10], while UTCS pre-trains temporal dynamics and subgraph knowledge [45]; PK-GCN applies a GNN to persistent  $k$ -core community search [13]. CS-TGN and UTCS do not explicitly model community cohesiveness, whereas PK-GCN is limited to persistent  $k$ -core structures. COCO instead couples interval-specific coreness computation with temporal learning: HM-Index predicts coreness for arbitrary query intervals, and AT-GNN uses it as a structural prior together with temporal interaction features. This coupling avoids materializing coreness for all  $O(t_{\max}^2)$  intervals and retraining the full GNN for each query.

### 3 Preliminary

A temporal graph is represented as  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ , where  $\mathcal{V}$  and  $\mathcal{E}$  denote the sets of vertices and temporal edges, respectively. Each edge  $e \in \mathcal{E}$  is a triplet  $(u, v, t)$ , representing an interaction between vertices  $u$  and  $v$  at timestamp  $t$ . The entire time interval of  $\mathcal{G}$  is  $\mathcal{T} = [1, t_{\max}]$ . For a given time interval  $[t_s, t_e] \subseteq \mathcal{T}$ , the temporal graph  $\mathcal{G}$  over  $[t_s, t_e]$  is denoted by  $\mathcal{G}_{[t_s, t_e]} = (\mathcal{V}_{[t_s, t_e]}, \mathcal{E}_{[t_s, t_e]})$ , where  $\mathcal{V}_{[t_s, t_e]} = \mathcal{V}$  and  $\mathcal{E}_{[t_s, t_e]} = \{(u, v, t) \mid (u, v, t) \in \mathcal{E}, t \in [t_s, t_e]\}$ . The projected graph of  $\mathcal{G}$  over  $[t_s, t_e]$  is denoted by  $\mathcal{G}_{pro[t_s, t_e]} = (\mathcal{V}_{pro[t_s, t_e]}, \mathcal{E}_{pro[t_s, t_e]})$ . Here,  $\mathcal{V}_{pro[t_s, t_e]} = \mathcal{V}$  and  $\mathcal{E}_{pro[t_s, t_e]} =$

$\{(u, v) \mid (u, v, t) \in \mathcal{E}, t \in [t_s, t_e]\}$ . We use  $N(v)$  to denote the neighbors of vertex  $v$  in  $\mathcal{G}$ . For the neighbors of  $v$  over a specific time interval  $[t_s, t_e]$ , we denote this set as  $N_{[t_s, t_e]}(v)$ . The degree of  $v$  at a particular timestamp  $t \in \mathcal{T}$  is denoted by  $d_t(v)$ . Additionally, the degree of  $v$  over the time interval  $[t_s, t_e]$  is  $d_{[t_s, t_e]}(v) = |N_{[t_s, t_e]}(v)|$ . Based on the above notations, we formally define our problem. For the query interval  $T_Q$ , we write  $N_{T_Q}(v) = \{u \in \mathcal{V} \mid \exists t \in T_Q : (u, v, t) \in \mathcal{E}\}$ . For a vertex set  $S$ ,  $N_{T_Q}(S) = (\bigcup_{v \in S} N_{T_Q}(v)) \setminus S$ .

**Problem 1. (Time-Interval-based Community Search (TICS)).** Given a temporal graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ , a time interval  $T_Q = [t_s, t_e]$ , and a set of query vertices  $V_Q \subseteq \mathcal{V}$ , the time-interval-based community search aims to find a vertex set  $C$  satisfying:

- (1)  $V_Q \subseteq C$ ;
- (2) The subgraph induced by  $C$  in  $\mathcal{G}_{[t_s, t_e]}$  is connected;
- (3) Vertices in  $C$  are densely intra-connected in  $\mathcal{G}_{[t_s, t_e]}$ .

Condition (3) requires both structural cohesiveness and frequent internal interactions. Temporal  $k$ -core constrains minimum degree in the projected graph, while QTCS [21] maximizes the minimum query-biased temporal degree based on temporal proximity to the query vertex. Next, we introduce several key concepts used in the rest of the paper.

**Definition 3.1. (Temporal  $k$ -core [43]).** Given a temporal graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  and a time interval  $[t_s, t_e]$ , let  $\mathcal{G}_{pro[t_s, t_e]}$  be the corresponding projected graph. The temporal  $k$ -core over  $[t_s, t_e]$  is the maximum subgraph  $\mathcal{H} \subseteq \mathcal{G}_{pro[t_s, t_e]}$  such that each vertex in  $\mathcal{H}$  has at least  $k$  neighbors.

**Definition 3.2. (Coreness).** Given a temporal graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  and a time interval  $[t_s, t_e]$ , the coreness of a vertex  $v \in \mathcal{V}$  over  $[t_s, t_e]$ , denoted by  $c_{[t_s, t_e]}(v)$ , is the largest  $k$  value such that  $v$  belongs to a temporal  $k$ -core in  $\mathcal{G}_{pro[t_s, t_e]}$ .

**Example 1.** Take Figure 1 as an example. Let the time interval be  $[3, 8]$ . The vertices  $v_6$  through  $v_{13}$  form a temporal 2-core. All these vertices have coreness 2, i.e.,  $c_{[3, 8]}(v_6) = \dots = c_{[3, 8]}(v_{13}) = 2$ .

**Graph Neural Networks.** GNNs [34] follow a neighborhood aggregation mechanism. This process can be expressed as:

$$\mathbf{h}_v^{(l)} = M \left( \mathbf{h}_v^{(l-1)}, \text{AGG} \left\{ \mathbf{h}_u^{(l-1)} : u \in N(v) \right\} \right) \quad (1)$$

Here,  $\mathbf{h}_v^{(l)}$  is the representation of vertex  $v$  in layer  $l$ , AGG is the aggregation function that combines messages from neighbors, and  $M$  is the message propagation function that updates the representation of the vertex using the aggregated representations and its own representation from the previous layer.

### 4 COCO Framework

To address the TICS problem, we present COCO, a cohesiveness-aware learning framework. COCO couples interval-specific coreness computation with temporal representation learning, using the predicted coreness as a structural prior. Compared to existing learning-based community search approaches, COCO is capable of identifying communities that demonstrate strong cohesiveness and can flexibly adapt to different graph structures across various queries.

Figure 2 shows COCO's three-stage pipeline: pre-training, fine-tuning, and searching. (1) In pre-training, the HM-Index learns

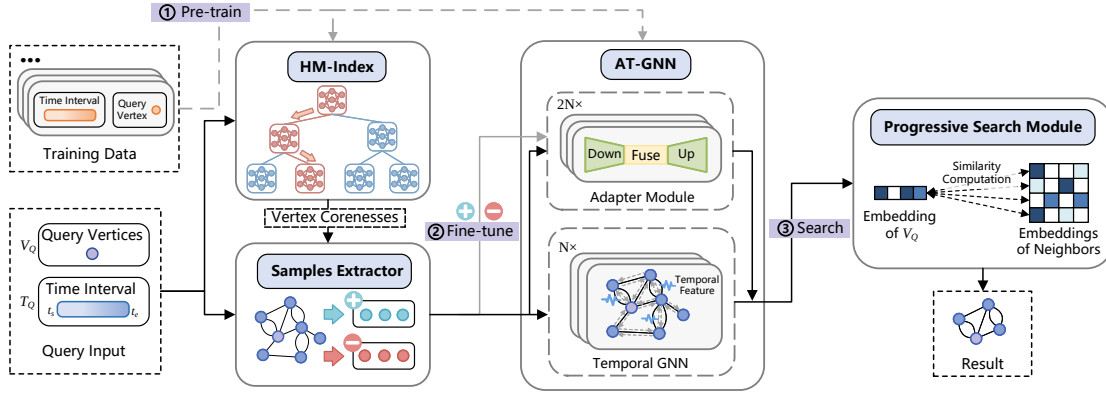


Figure 2: The design of COCO framework

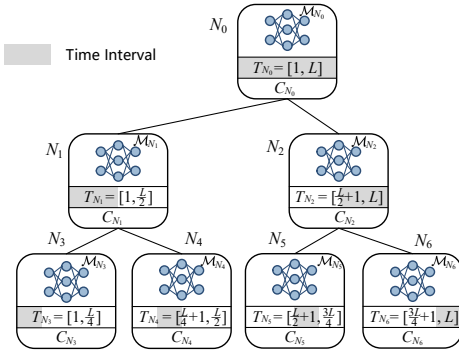


Figure 3: Architecture of HM-Index

to predict vertex coreness over time, while the AT-GNN captures common temporal community patterns. (2) During fine-tuning, the HM-Index predicts coreness for query vertices  $V_Q$  and their neighbors in the time interval  $T_Q$ . A sample extractor then creates positive and negative examples based on these coreness values and the query context, used to fine-tune the AT-GNN's adapter module. (3) In the search phase, AT-GNN generates vertex embeddings. The progressive search module iteratively selects vertices most similar to  $V_Q$ , adding them to the community until constraints are met. Due to space limitations, some proofs are provided in the **technical report [1]**.

This section defines the architectures and inference procedures of HM-Index and AT-GNN, while Section 5 describes how they are constructed, trained, and used. The interval-partition rule below specifies the HM-Index structure, and Algorithm 3 applies this rule during construction.

#### 4.1 Hierarchical Model Index

To integrate coreness with GNNs, we need to compute coreness for arbitrary query intervals  $T_Q$ , since different intervals may produce dramatically different graph topologies. Computing coreness dynamically for each query is accurate but costly, while precomputing coreness for all  $O(t_{\max}^2)$  possible subintervals requires excessive storage and preprocessing time [44]. This presents a fundamental trade-off between efficiency and flexibility.

We propose a learning-based method that predicts coreness instead of computing it directly. Our approach divides the entire time

span into smaller intervals and uses lightweight regression models to predict coreness within each. These models are organized into a Hierarchical Model Index (HM-Index) over recursively nested time intervals, as illustrated in Figure 3. Each node  $N$  in the HM-Index has three components: (1)  $T_N$ : the time interval it covers; (2)  $M_N$ : a lightweight model that predicts vertex coreness for any subinterval of  $T_N$ ; (3)  $C_N = \{c_{T_N}(v) \mid v \in \mathcal{V}\}$ : the set of vertex coreness values over  $T_N$ , used by the parent node to improve prediction accuracy. This nested organization lets each child model reuse the parent prediction and refine it with information from a narrower interval. This residual refinement design is formalized in Lemma 4.1.

**Time Interval Partition Rules.** The HM-Index employs a recursive partitioning scheme: the root node spans the entire temporal range  $[1, t_{\max}]$ ; each parent node divides its interval into  $p$  equal subintervals; partitioning terminates when leaf node's time interval length falls below the given threshold  $x$ . For example, in Figure 3,  $p = 2$  and  $x = \frac{L}{4}$ . The HM-Index structure maintains two key properties: (1) *Strict Hierarchy*: for any child node  $N'$  of a node  $N$ , we have  $T_{N'} \subset T_N$ . (2) *Disjoint Siblings*: Nodes at the same depth have non-overlapping time intervals.

**Model Structure.** The regression model  $M_N$  in a node  $N$  is a lightweight Multilayer Perceptron (MLP). It consists of four components: (1) A linear transformation layer  $W_0 \in \mathbb{R}^{\dim_h \times 1}$  for single-valued features; (2) A timestamp feature processing layer  $W_1 \in \mathbb{R}^{\dim_h \times 2 \lceil \frac{|T_N|}{p} \rceil}$ ; (3) A time interval feature processing layer  $W_2 \in \mathbb{R}^{\dim_h \times p}$ ; (4) A feature fusion layer  $W_f \in \mathbb{R}^{1 \times 3 \cdot \dim_h}$  that merges the three processed features into a unified representation. Here,  $\dim_h$  represents the hidden dimension. The model processes different feature types separately and then fuses them, allowing specialized feature extraction while producing a more comprehensive representation.

**Coreness Prediction Algorithm.** The coreness prediction algorithm uses a multi-stage approach that traverses the HM-Index from the root, refining predictions at each level. This top-down traversal passes each prediction to the selected child model, which refines it using more localized interval information. We first describe the input to the model  $M_N$ , then present the full algorithm that coordinates this hierarchical prediction process.

Given an HM-Index node  $N$ , a vertex  $v$ , and a query time interval  $[t_s, t_e] \subseteq T_N$ , the interval may span multiple child nodes of  $N$ . Let

**Algorithm 1: Coreness Prediction Algorithm**


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**Input:** A temporal graph  $\mathcal{G}$ , a vertex  $v$ , a query time interval  $[t_s, t_e]$ , HM-Index root node  $N_{root}$   
**Output:** Predicted coreness  $\hat{c}$  for  $v$  over  $[t_s, t_e]$

```

1  $N \leftarrow N_{root}; \hat{c} \leftarrow 0;$ 
2 while True do
3    $s_1 \leftarrow [c_t(v) || d_t(v)]_{t=t_s, t \notin [t'_s, t'_e]}$ ;
4    $s_2 \leftarrow [C_{T_{N_i}} || d_{T_{N_i}}(v)]_{i=0}^{N_c-1}$ ;
5    $\hat{c} \leftarrow \mathcal{M}_N(\hat{c}, s_1, s_2);$  // Eq. (2)
6   for  $N' \in N.child$  do
7     if  $[t_s, t_e] \subseteq T_{N'}$  then
8        $N \leftarrow N';$  break;
9   if no child node contains  $[t_s, t_e]$  then break;
10 return  $\hat{c}$ ;
```

---

$N_c = \{N_0, N_1, \dots, N_{|N_c|-1}\}$  be the ordered sequence of these child nodes, covering a combined interval  $[t'_s, t'_e]$ . Let  $N_p$  be the parent of  $N$ . The model  $\mathcal{M}_N$  predicts the coreness of  $v$  as follows.

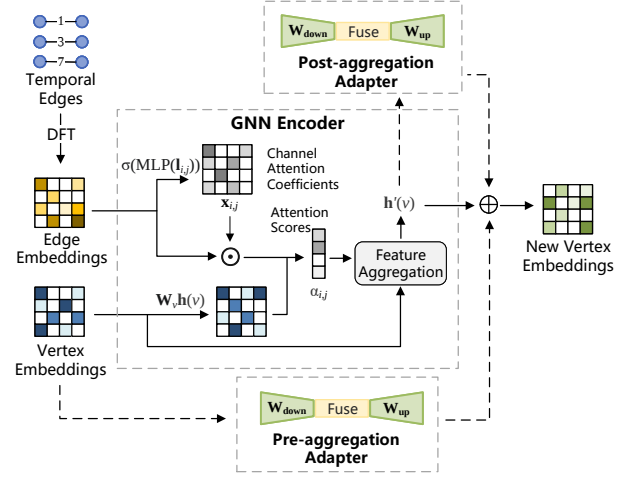
$$\hat{c}_{[t_s, t_e]}^N(v) = \mathcal{M}_N(\hat{c}, s_1, s_2) = \mathbf{W}_f(\mathbf{W}_0 \hat{c} || \mathbf{W}_1 s_1 || \mathbf{W}_2 s_2) \quad (2)$$

The model  $\mathcal{M}_N$  takes three inputs to predict the coreness of a vertex  $v$ :  $\hat{c}$ ,  $s_1$ , and  $s_2$ .  $\hat{c}$  is the coreness value predicted by the parent node  $N_p$ , or 0 if no parent prediction exists. It is used as the current estimate for  $\mathcal{M}_N$ , while  $s_1$  and  $s_2$  provide interval-specific information for refining that estimate. The vector  $s_1$  encodes the coreness and degree of  $v$  at timestamps outside a given interval  $[t'_s, t'_e]$ , capturing external temporal dynamics. The vector  $s_2$  aggregates  $v$ 's coreness and degree across all time intervals in the child nodes  $N_c$ , providing a summary of its structural properties over time.

Based on the coreness prediction of a single HM-Index node, the complete coreness prediction algorithm based on the HM-Index is outlined in Algorithm 1. It starts by setting the root node  $N$  and initializing the predicted coreness  $\hat{c}$  to 0 (line 1). Then, Algorithm 1 traverses the HM-Index as follows. For the currently visited HM-Index node  $N$ , two feature vectors  $s_1$  and  $s_2$  are computed (lines 3–4), based on which the coreness  $\hat{c}$  is updated (line 5). Next, Algorithm 1 checks if there exists a child node  $N'$  of  $N$  that fully covers the query time interval  $[t_s, t_e]$  (lines 6–8). If so, Algorithm 1 visits  $N'$  and passes the updated estimate  $\hat{c}$  to the child model for further refinement. Otherwise, Algorithm 1 terminates. This root-to-leaf update implements the residual refinement process analyzed in Lemma 4.1.

**Theoretical Error Analysis.** We model HM-Index's hierarchical prediction as a sequential refinement process. During the root-to-leaf traversal in Algorithm 1, each node uses increasingly localized temporal features to correct its parent's prediction. This formulation separates the inherited prediction error from the local approximation error introduced at each level. The following lemma gives an error bound for this top-down residual learning under the stated contraction condition. The contraction rate  $\beta < 1$  is treated as an assumption of this refinement process and is evaluated empirically in Appendix A. Let  $c_{[t_s, t_e]}^*(v)$  denote the exact coreness of  $v$  over the query interval  $[t_s, t_e]$ .

**LEMMA 4.1.** Assume each model  $\mathcal{M}_{N_j}$  at depth  $j$  in Algorithm 1 satisfies the contraction property  $\epsilon_j \leq \beta \epsilon_{j-1} + \eta$ , where  $\beta \in [0, 1)$  is the contraction rate,  $\eta \geq 0$  is the irreducible local error, and  $\epsilon_j =$



**Figure 4: Architecture of AT-GNN**

$|\hat{c}^{(j)} - c_{[t_s, t_e]}^*(v)|$ . The final prediction error after traversing  $h$  levels is strictly bounded by  $\epsilon_h \leq \beta^h \epsilon_0 + \frac{\eta(1-\beta^h)}{1-\beta}$ .

**PROOF.** HM-Index operates as a coarse-to-fine residual chain. At depth  $j$ ,  $\mathcal{M}_{N_j}$  treats the parent prediction  $\hat{c}^{(j-1)}$  as a structural prior, utilizing local temporal features to learn a residual correction. Here,  $\beta < 1$  is the contraction assumption, while  $\eta$  bounds the remaining local approximation error.

We prove  $\epsilon_j \leq \beta^j \epsilon_0 + \eta \sum_{k=0}^{j-1} \beta^k$  by induction. The base case  $j = 0$  is trivial. Assuming this holds for depth  $j$ , the contraction property yields  $\epsilon_{j+1} \leq \beta \epsilon_j + \eta \leq \beta(\beta^j \epsilon_0 + \eta \sum_{k=0}^{j-1} \beta^k) + \eta = \beta^{j+1} \epsilon_0 + \eta \sum_{k=0}^j \beta^k$ , completing the induction.

Evaluating at the final depth  $h$  and computing the geometric series yields  $\epsilon_h \leq \beta^h \epsilon_0 + \frac{\eta(1-\beta^h)}{1-\beta}$ . Thus, the initial propagated error  $\epsilon_0$  decays exponentially with depth, while the cumulative local error is bounded by  $\frac{\eta}{1-\beta}$ . Under the stated contraction condition, this bound prevents error explosion.  $\square$

The quantity  $\eta/(1-\beta)$  is a conservative upper bound obtained from a uniform local error  $\eta$ . Appendix A discusses the contraction assumption, the risk of overfitting at deeper nodes, and the tightness of this bound, and reports empirical values of  $\beta$  and the per-level approximation errors  $\eta_k$ .

**Complexity Analysis.** Let  $t_{\max}$  be the maximum timestamp of the graph, and  $x$  be the maximum length of the time interval for the leaf nodes in the HM-Index. The number of leaf nodes in the HM-Index is bounded by  $O(\frac{t_{\max}}{x})$ . Since the HM-Index is a balanced tree, the total number of nodes is also bounded by  $O(\frac{t_{\max}}{x})$ . The height of the HM-Index is  $h = \log_p \frac{t_{\max}}{x}$ . In the worst case, Algorithm 1 traverses the HM-Index from the root to the leaf node. Hence, the time complexity of Algorithm 1 is bounded by  $O(\log \frac{t_{\max}}{x})$ . Each node contains a model and a set of coreness values for vertices over the corresponding time interval. Therefore, the space required for a single node is bounded by  $O(|M| + |\mathcal{V}|)$ . Consequently, the overall space complexity of HM-Index is bounded by  $O(\frac{t_{\max}}{x} \cdot (|M| + |\mathcal{V}|))$ .

## 4.2 Adaptive Temporal GNN

The Adaptive Temporal Graph Neural Network (AT-GNN) is designed to capture both the structural and temporal characteristics of a temporal graph while supporting efficient fine-tuning for a variety of TICS queries.

Figure 4 shows the architecture of one layer in the AT-GNN. The AT-GNN contains  $K$  layers. The model has two main components: the encoder and the adapter. Each layer includes two adapters, namely the *Pre-aggregation Adapter* and the *Post-aggregation Adapter*. The vertex embedding  $h^{(l)}(v)$  in the  $l$ -th layer of the AT-GNN is computed as follows.

$$h^{(l)}(v) = \text{ENC}(h^{(l-1)}(v)) + \text{ADP}_{\text{pre}}(h^{(l-1)}(v)) + \text{ADP}_{\text{post}}(\text{ENC}(h^{(l-1)}(v))) \quad (3)$$

Here,  $\text{ENC}(\cdot)$  is the encoder,  $\text{ADP}_{\text{pre}}(\cdot)$  and  $\text{ADP}_{\text{post}}(\cdot)$  are pre-aggregation and post-aggregation adapters, respectively. Next, we introduce them in detail.

**Encoder.** For a vertex  $v$ , the initial feature  $h^{(0)}(v) \in \mathbb{R}^{\dim_v}$  is defined as:

$$h^{(0)}(v) = f_v([q(v) || s_c(v)]), \quad (4)$$

where  $q(v)$  indicates whether  $v$  is a query vertex (1 if  $v \in V_Q$ , else 0), and  $s_c(v) = [c_t(v)]_{t=t_s}^{t_e}$  is the temporal coreness sequence. The function  $f_v(\cdot)$  adjusts the dimension via linear interpolation (for short sequences) or average pooling (for long ones). **Vertex features are constructed in the original domain from  $q(v)$  and the normalized coreness sequence  $s_c(v)$ . Temporal interaction patterns are encoded from the time-varying edge-presence sequence using DFT.** For an edge  $(v, v')$ , the initial embedding  $\mathbf{l}_{v,v'} \in \mathbb{R}^{\dim_e}$  is

$$\mathbf{l}_{v,v'} = [|\text{DFT}(f_e([a_t(v, v')])_{t=t_s}^{t_e})|]_{0:\dim_e}. \quad (5)$$

The edge presence sequence  $[a_t(v, v')]_{t=t_s}^{t_e}$  indicates whether edge  $(v, v')$  exists at time  $t$ , where  $a_t(v, v') = 1$  if the edge exists and 0 otherwise. The function  $f_e(\cdot)$  performs dimension transformation, similar to  $f_v(\cdot)$ .

**LEMMA 4.2.** Let  $\mathbf{a}, \mathbf{b} \in \{0, 1\}^T$  be two edge presence sequences with Hamming distance  $\|\mathbf{a} - \mathbf{b}\|_0 = m$ , and let  $\phi_d(\mathbf{a}) = [\hat{a}_k]_{k=0}^{d-1}$  denote the truncated DFT amplitude embedding defined in Eq. (5) with  $d \leq T$ . Then  $\|\phi_d(\mathbf{a}) - \phi_d(\mathbf{b})\|_2 \leq \sqrt{m}$ .

The  $\sqrt{m}$  bound shows that small changes in an edge-presence sequence produce bounded changes in its DFT amplitude embedding. This provides a stability property for the temporal edge representation before it is used by channel attention and neighbor aggregation. After applying DFT, the first  $\dim_e$  elements of the amplitude vector form the edge embedding. To strengthen temporal modeling, a channel attention mechanism refines the embeddings:

$$\mathbf{x}_{v,v'} = \sigma(\text{MLP}(\mathbf{l}_{v,v'})), \quad \mathbf{l}'_{v,v'} = \mathbf{l}_{v,v'} \odot \mathbf{x}_{v,v'}. \quad (6)$$

The coefficient vector  $\mathbf{x}_{v,v'}$  is computed from the DFT amplitude embedding  $\mathbf{l}_{v,v'}$  of the current edge and serves as an input-dependent channel gate. Through element-wise multiplication, it rescales the original frequency-channel amplitudes while preserving their coordinates before neighbor aggregation.

For vertex aggregation, the encoder calculates attention scores  $b_{v,v'}$  by combining vertex and edge features:

### Algorithm 2: Forward Propagation of AT-GNN

---

**Input:** A temporal graph  $\mathcal{G}$ , a set of query vertices  $V_Q$ , a query time interval  $T_Q$   
**Output:** The embeddings of  $\mathcal{V}$

- 1  $\{h^{(0)}(v) \leftarrow f_v([q(v) || s_c(v)]) \mid v \in \mathcal{V}\}$  // Eq. (4)
- 2  $\{\mathbf{l}_{v,v'} \leftarrow [|\text{DFT}(f_e([a_t(v, v')])_{t=t_s}^{t_e})|]_{0:\dim_e} \mid e = (v, v'), t \in \mathcal{E}, t \in T_Q\}$  // Eq. (5)
- 3 **for**  $l = 1$  **to**  $K$  **do**
- 4   **for**  $v \in \mathcal{V}$  **do**
- 5      $h_{\text{pre}}(v) \leftarrow \text{ADP}(h^{(l-1)}(v))$ ; // Pre-aggregation adapter
- 6     // encoder Eq. (6) to (8)
- 7     **for**  $v' \in \{v\} \cup N_{T_Q}(v)$  **do**
- 8        $\mathbf{x}_{v,v'} \leftarrow \sigma(\text{MLP}(\mathbf{l}_{v,v'})), \mathbf{l}'_{v,v'} \leftarrow \mathbf{l}_{v,v'} \odot \mathbf{x}_{v,v'}$ ;
- 9        $b_{v,v'} = \text{LeakyReLU}(\mathbf{a}^T \cdot [\mathbf{W}_v h^{(l-1)}(v) || \mathbf{W}_v h^{(l-1)}(v') || \mathbf{l}'_{v,v'}])$ ;
- 10        $\alpha_{v,v'} = \text{softmax}_{v'}(b_{v,v'})$ ;
- 11        $h'(v) \leftarrow \sum_{v' \in N(v) \cup \{v\}} \alpha_{v,v'} \cdot h^{(l-1)}(v')$ ;
- 12        $h_{\text{post}}(v) \leftarrow \text{ADP}(h'(v))$ ; // Post-aggregation adapter
- 13      $h^{(l)}(v) \leftarrow h_{\text{pre}}(v) + h'(v) + h_{\text{post}}(v)$ ;
- 13 **return**  $\{h^{(l)}(v) \mid v \in \mathcal{V}\}$ ;

**Procedure**  $\text{ADP}(h_{\text{in}}(v))$

- 14  $s_Q \leftarrow [\bar{c}(V_Q) || \bar{c}(N(V_Q))]_{\frac{t_s}{t_{\max}}}^{\frac{t_e}{t_{\max}}}$ ;
- 15  $\tilde{\mathbf{h}}(v) \leftarrow [\mathbf{W}_{\text{down}_1} h_{\text{in}}(v) || \mathbf{W}_{\text{down}_2} s_Q]$ ; // Eq. (9)
- 16  $\mathbf{q} \leftarrow \mathbf{W}_q \tilde{\mathbf{h}}(v), \mathbf{k} \leftarrow \mathbf{W}_k \tilde{\mathbf{h}}(v), \mathbf{v} \leftarrow \mathbf{W}_v \tilde{\mathbf{h}}(v)$ ; // Eq. (9)
- 17  $\alpha \leftarrow \text{softmax}(\frac{\mathbf{q}^T \mathbf{k}}{\sqrt{\dim_{\text{head}}}})$ ; // Eq. (10)
- 18 **return**  $\text{ReLU}(\mathbf{W}_{\text{up}}(\alpha \cdot \mathbf{v}))$ ; // Eq. (10)

---

$$b_{v,v'} = \text{LeakyReLU}(\mathbf{a}^T \cdot [\mathbf{W}_v h^{(l-1)}(v) || \mathbf{W}_v h^{(l-1)}(v') || \mathbf{l}'_{v,v'}]). \quad (7)$$

Here,  $\mathbf{a}$  is a learnable vector and  $\mathbf{W}_v$  projects vertex features. Dynamic edge features  $\mathbf{l}'_{v,v'}$  capture temporal interactions. The normalized attention scores are then used to aggregate neighboring features to update  $h^{(l)}(v)$ , as in Eq. (8).

$$\alpha_{v,v'} = \text{softmax}_{v'}(b_{v,v'}) = \frac{\exp(b_{v,v'})}{\sum_{v'' \in N(v) \cup \{v\}} \exp(b_{v,v''})}, \quad (8)$$

$$\text{ENC}(h^{(l-1)}(v)) = \sum_{v' \in N(v) \cup \{v\}} \alpha_{v,v'} \cdot h^{(l-1)}(v')$$

**Adapter.** COCO is designed to handle TICS queries under varying graph structures. Fully retraining or fine-tuning the encoder for different queries would be prohibitively expensive. To address this issue, we design a query context-aware adapter. This lightweight trainable module is inserted into each encoder layer, both before and after the aggregation step. This dual-adapter architecture improves generalization by enabling controlled feature adaptation while preserving the original graph representations.

Each adapter includes three parts: dimension reduction, feature fusion, and dimension expansion. In the dimension reduction stage, two inputs are processed. First, vertex embeddings  $h_{\text{in}}(v)$  are linearly projected to a lower dimension using  $\mathbf{W}_{\text{down}_1}$ . Second, query context features are extracted: average coreness of query vertices and their neighbors ( $\bar{c}(V_Q)$  and  $\bar{c}(N(V_Q))$ ), and normalized time features  $\frac{t_s}{t_{\max}}$  and  $\frac{|T_Q|}{t_{\max}}$ . These features are combined into a vector  $s_Q$ , then projected via  $\mathbf{W}_{\text{down}_2}$ . After concatenation, three learnable

matrices  $\mathbf{W}_q$ ,  $\mathbf{W}_k$ , and  $\mathbf{W}_v$  project the input into query, key, and value vectors, as shown in Eq. (9).

$$\begin{aligned}\tilde{\mathbf{h}}(v) &= [\mathbf{W}_{\text{down}_1} \mathbf{h}_{in}(v) \| \mathbf{W}_{\text{down}_2} \mathbf{s}_Q], \\ \mathbf{q} &= \mathbf{W}_q \tilde{\mathbf{h}}(v), \quad \mathbf{k} = \mathbf{W}_k \tilde{\mathbf{h}}(v), \quad \mathbf{v} = \mathbf{W}_v \tilde{\mathbf{h}}(v)\end{aligned}\quad (9)$$

In the final dimension expansion stage, the output is further transformed by a learnable matrix  $\mathbf{W}_{\text{up}}$ . The complete adapter output for input  $\mathbf{h}_{in}(v)$  is given in Eq. (10):

$$\alpha = \text{softmax}\left(\frac{\mathbf{q}^\top \mathbf{k}}{\sqrt{\text{dim}_{\text{head}}}}\right), \quad \text{ADP}(\mathbf{h}_{in}(v)) = \text{ReLU}(\mathbf{W}_{\text{up}}(\alpha \cdot \mathbf{v})) \quad (10)$$

This reduce-then-expand design limits the dimension of the adapter update while retaining query-specific adjustments.

**LEMMA 4.3.** *For a fixed query context  $\mathbf{s}_Q$ , a fixed active region of the ReLU, and a linear/identity configuration of the fusion module, the adapter update has jacobian rank at most  $\xi = \min\{\text{dim}_{\text{ADP}}, \text{dim}_{\text{head}}\}$ . Within this region, it can exactly realize any affine correction  $\Delta_Q(\mathbf{h}) = \mathbf{A}_Q \mathbf{h} + \mathbf{b}_Q$  satisfying  $\text{rank}([\mathbf{A}_Q \ \mathbf{b}_Q]) \leq \xi$ , provided that its output remains componentwise nonnegative.*

Here,  $\text{dim}_{\text{ADP}}$  denotes the hidden dimension of the adapter's bottleneck, and  $\text{dim}_{\text{head}}$  is the dimension of the attention head. Lemma 4.3 first bounds the rank of the update produced through the down-projection and up-projection. Within a fixed activation region, its exact-realization result characterizes the query-specific affine corrections represented by the adapter. The bottleneck dimension therefore controls the capacity of query-specific adaptation while keeping the pre- and post-aggregation adapters compact.

**Forward Propagation.** As shown in Algorithm 2, forward propagation of AT-GNN begins by computing initial embeddings for vertices and edges (lines 1–2). Then, the model performs  $K$  layers of message passing. Each layer includes a pre-aggregation adapter, an encoder, and a post-aggregation adapter. In the pre-aggregation adapter, the input embedding  $\mathbf{h}^{(l-1)}(v)$  is transformed into an intermediate representation  $\mathbf{h}_{\text{pre}}(v)$  using the ADP procedure (line 5). ADP improves embeddings by adding query-specific context. It builds a context vector  $\mathbf{s}_Q$  from the query vertex set  $V_Q$  and its neighbors, applies self-attention, and outputs a refined representation (lines 14–18). Next, the encoder aggregates neighbor information. For each neighbor  $v'$  of  $v$ , attention scores  $\alpha_{v,v'}$  are calculated using channel attention, and used to combine neighbor embeddings (lines 6–10). The result,  $\mathbf{h}'(v)$ , is further processed by the post-aggregation adapter via ADP (line 11). Finally, the output combines features from all three components (line 12).

## 5 COCO Workflow

In this section, we present the workflow of COCO, which includes pre-training, fine-tuning, and searching phases. Algorithm 3 constructs HM-Index according to the interval-partition rule in Section 4.1, while the remaining procedures train the components and use them for query processing.

### 5.1 Pre-training Phase

In the pre-training phase, COCO builds the HM-Index and trains the AT-GNN. Next, we provide a detailed explanation of training for these two components.

**Construction of the HM-Index.** The HM-Index construction algorithm builds a hierarchical tree structure to predict vertex coreness in temporal graphs. Using a top-down strategy, the algorithm recursively divides time intervals into smaller segments. For each resulting time interval, it creates a corresponding tree node and trains a specialized prediction model.

As outlined in Algorithm 3, the procedure starts by initializing a queue  $L$  containing the root node. The root node covers the entire time interval  $\mathcal{T}$  (line 1). For each node  $N$  from the queue, the algorithm first creates a model (line 4) and then randomly samples batches of vertices (line 5). For each vertex  $v$  in the batch, it selects a random subinterval of  $T_N$  (line 9). The coreness of  $v$  over this subinterval is computed using the core decomposition method [14] and serves as the ground-truth label (line 10). Meanwhile, Algorithm 1 generates the predicted coreness of  $v$  (line 11). The training objective is defined as the mean squared error between the predicted and actual values.

$$\mathcal{L}_c = \frac{1}{n_b} \sum_{i=1}^{n_b} (\hat{c}(v_i) - c(v_i))^2 \quad (11)$$

This loss is calculated for each batch (line 12), and the parameters of the model  $\mathcal{M}_N$  are updated accordingly (line 13). If the current node's time interval has a length greater than or equal to  $x$ , it is divided into  $p$  subintervals (line 15), and child nodes corresponding to these new intervals are enqueued (line 16). This iterative process continues until all eligible nodes are processed, after which the HM-Index is constructed.

**Training for AT-GNN.** Most learning-based community search methods depend on ground-truth communities. However, this approach does not work for temporal graphs because there are no ground-truth communities available for existing temporal graphs [21]. To address this, we propose a contrastive learning method to train AT-GNN in an unsupervised manner. We design two loss functions for training AT-GNN: the cohesive loss and the temporal link loss.

The cohesive loss function is designed to bring the embeddings of potentially related vertices (likely belonging to the same community) closer together in the latent space, while simultaneously pushing unrelated vertices farther apart. To select positive samples, we define a cohesiveness score  $\text{CoS}(v)$  for each candidate vertex  $v$  over  $T_Q$  as follows.

$$\text{CoS}(v) = c(v) \cdot \frac{|\{e = (v, u, t) | u \in N(v), t \in T_Q\}|}{|N_{T_Q}(v)|} \quad (12)$$

Here,  $c(v)$  is the coreness of  $v$ , while the second factor is its average number of temporal interactions per active neighbor in  $T_Q$ . Their product scores candidates by local structural and temporal cohesiveness.

**Sampling Rule 1.** To enable contrastive learning in AT-GNN, positive and negative samples are generated for the cohesive loss.

(1) **Positive Samples  $V_{\text{pos}}$ :** Starting from  $V_Q$ ,  $V_{\text{pos}}$  is expanded by repeatedly adding the vertex  $v'$  with the highest  $\text{CoS}(v')$  from  $N_{T_Q}(V_{\text{pos}}) \setminus V_{\text{pos}}$  until the number of positive samples reaches  $|V_{\text{smp}}|$ . This frontier expansion keeps  $V_{\text{pos}}$  connected to  $V_Q$ .

(2) **Negative Samples  $V_{\text{neg}}$ :**  $V_{\text{neg}}$  is formed by randomly selecting  $|V_{\text{smp}}|$  vertices from  $\{\mathcal{V} \setminus V_{\text{pos}}\}$ .

**Algorithm 3: HM-Index Construction**


---

**Input:** A temporal graph  $\mathcal{G}$ , a partition coefficient  $p$ , a length threshold  $x$ , a learning rate  $\varphi$ , a number of epochs  $\lambda$ , a batch size  $n_b$

**Output:** Root node of HM-Index

```

1 Initialize queue  $L$  with a node  $N_{root}$  with entire time interval  $\mathcal{T}$ ;
2 while  $L \neq \emptyset$  do
3    $N \leftarrow L.dequeue()$ ; initialize a model  $\mathcal{M}_N$  for  $N$ ;
4   Sample batches  $\{V_1, V_2, \dots, V_n\}$  where each  $|V_{b_i}| = n_b$ ;
5   for  $epoch = 1$  to  $\lambda$  do
6     for each batch  $V_i \in \{V_1, V_2, \dots, V_n\}$  do
7        $\mathcal{L}_c \leftarrow 0$ ;
8       for  $v \in V_i$  do
9          $[t_s, t_e] \leftarrow$  randomly sample subinterval from  $T_N$ ;
10         $c(v)_{[t_s, t_e]} \leftarrow$  compute the coreness of  $v$ ;
11         $\hat{c}_{[t_s, t_e]}(v) \leftarrow$  predict using Algorithm 1;
12         $\mathcal{L}_c \leftarrow \mathcal{L}_c + \frac{1}{n_b} (\hat{c}(v) - c(v))^2$ ;
13       $\theta(\mathcal{M}_N) \leftarrow \theta(\mathcal{M}_N) - \varphi \cdot \nabla_{\theta} \mathcal{L}_c$ ;
14  if  $|T_N| > x$  then
15    Create  $p$  child nodes with equal-length subintervals;
16    Enqueue child nodes into  $L$ ;
17 return  $N_{root}$ ;

```

---

Once the positive and negative samples are identified, we model the cohesive loss using the margin triplet loss [31]. It minimizes the distance between query vertices and positive samples while enforcing a margin between the query and negative samples.

$$\mathcal{L}_{Coh} = \frac{1}{|V_{pos}| \cdot |V_{neg}|} \cdot \sum_{u \in V_{pos}} \sum_{v \in V_{neg}} \max(0, dis(u, V_Q) - dis(v, V_Q) + \epsilon) \quad (13)$$

Here,  $dis(\cdot)$  computes the average L2 distance between a vertex and query vertices. For example,  $dis(v, V_Q) = \frac{1}{|V_Q|} \cdot \sum_{v' \in V_Q} \|h(v) - h(v')\|$ .  $\epsilon$  is the margin value.

A vertex may have a different number of connections with its neighbors. The idea behind the temporal link loss is to enhance the similarity of vertices with frequent connections while distinguishing weakly linked vertices. To measure the strength of the connection between two vertices, we define a function of link strength.

$$E(v, u) = \{e = (v, u, t) | e \in \mathcal{E}, t \in T_Q\},$$

$$LinkS(v, u) = \frac{|E(v, u)|}{\max_{(v, u, t) \in E(v, u)} t - \min_{(v, u, t) \in E(v, u)} t + 1} \quad (14)$$

**Sampling Rule 2.** The samples for the temporal link loss are triples, denoted as  $V_{temp}$ . Each triple  $(v, v_p, v_n) \in V_{temp}$  consists of an anchor vertex, a positive vertex, and a negative vertex.

- (1) Anchor Vertex  $v$ :  $v$  is randomly selected from  $\mathcal{V}$ .
- (2) Positive Vertex  $v_p$ : Among the neighbors of  $v$ , the positive vertex is  $v_p = \arg \max_{v' \in N_{T_Q}(v)} LinkS(v, v')$ .
- (3) Negative Vertex  $v_n$ : Among the neighbors of  $v$ , the negative vertex is  $v_n = \arg \min_{v' \in N_{T_Q}(v)} LinkS(v, v')$ .

Based on the sampled triples, the temporal link loss is defined in Eq.(15). Finally, we combine the cohesive loss and the temporal link loss as shown in Eq.(16).

$$\mathcal{L}_{Temp} = \frac{1}{|V_{temp}|} \cdot \sum_{(v_a, v_p, v_n) \in V_{temp}} \max(0, dis(v_a, v_p) - dis(v_a, v_n) + \epsilon) \quad (15)$$

**Algorithm 4: AT-GNN Pre-training**


---

**Input:** A temporal graph  $\mathcal{G}$ , a learning rate  $\varphi$ , a number of epochs  $\lambda$ , a batch size  $n_b$ , an initialized AT-GNN model  $\mathcal{M}_{AT}$

**Output:** The trained model  $\mathcal{M}_{AT}$

```

1 Generate query batches  $\{Q_1, Q_2, \dots, Q_n\}$  where  $Q_i$  has  $n_b$  queries;
2 for  $epoch = 1$  to  $\lambda$  do
3   for each batch  $Q_i \in \{Q_1, Q_2, \dots, Q_n\}$  do
4      $\mathcal{L} \leftarrow 0$ ;
5     for each query  $(V_Q, T_Q) \in Q_i$  do
6        $\{h^{(l)}(v) | v \in \mathcal{V}\} \leftarrow$  Algorithm 2;
7        $V_{pos}, V_{neg} \leftarrow$  sample by rule 1;
8        $\mathcal{L}_{Coh} \leftarrow \frac{1}{n_b} \cdot \frac{1}{|V_{pos}| \cdot |V_{neg}|} \sum_{u \in V_{pos}} \sum_{v \in V_{neg}} \max(0, dis(u, V_Q) - dis(v, V_Q) + \epsilon)$ ;
9        $V_{temp} \leftarrow$  sample by rule 2;
10       $\mathcal{L}_{Temp} \leftarrow \frac{1}{n_b} \cdot \frac{1}{|V_{temp}|} \sum_{(v_a, v_p, v_n) \in V_{temp}} \max(0, dis(v_a, v_p) - dis(v_a, v_n) + \epsilon)$ ;
11       $\mathcal{L} \leftarrow \mathcal{L}_{Coh} + \delta \mathcal{L}_{Temp}$ ;
12     $\theta(\mathcal{M}_{AT}) \leftarrow \theta(\mathcal{M}_{AT}) - \varphi \cdot \nabla_{\theta} \mathcal{L}$ ;
13 return  $\mathcal{M}_{AT}$ ;

```

---

$$\mathcal{L} = \mathcal{L}_{Coh} + \delta \mathcal{L}_{Temp} \quad (16)$$

The overall training process of AT-GNN is presented in Algorithm 4. The algorithm starts by generating several query batches (line 1). Each query has a different query vertex and a different time interval. This allows AT-GNN to learn community information across various time intervals. For each query, the algorithm samples data for the cohesive loss and computes the loss (lines 7-8). It then samples data for the temporal link loss and computes the corresponding loss (lines 9-10). These two losses are combined (line 11). After processing each batch, the parameters of AT-GNN are updated (line 12).

**5.2 Fine-tuning Phase**

In the fine-tuning phase, the adapter module in AT-GNN plays a key role. When a query arrives, COCO avoids heavy query overhead by fine-tuning only the parameters in the adapter module. The process of fine-tuning is shown in Algorithm 5.

First, COCO predicts the coreness of all vertices using HM-Index via Algorithm 1 (line 1). Then it samples vertices based on Rule 1 and Rule 2 from Section 5.1 (lines 2). For each fine-tuning step, COCO generates the vertex embeddings using AT-GNN (line 4). Next, sampled vertices are used to compute the loss (line 5). The key difference between pre-training and fine-tuning lies in parameter updates. During pre-training, all parameters in AT-GNN are updated. However, during fine-tuning, the parameters in the GNN encoder are frozen. Only the parameters in the adapter module are updated based on the computed loss (line 6).

This approach offers several benefits. First, freezing the GNN encoder reduces computational costs significantly. Since the encoder parameters remain unchanged, the model avoids redundant updates, making the fine-tuning process more efficient. Second, fine-tuning only the adapter module allows the model to adapt quickly to new queries while retaining the general knowledge learned during pre-training. This ensures that the model remains robust and accurate even with limited query-specific data.

**Algorithm 5: COCO Fine Tuning & Searching**

**Input:** A temporal graph  $\mathcal{G}$ , a set of query vertices  $V_Q$ , a query time interval  $T_Q$ , a learning rate  $\varphi$ , a number of fine tuning  $\lambda$ , the AT-GNN  $\mathcal{M}_{AT}$  model, adapter modules  $\mathcal{M}_{ADP}$ , HM-Index root node  $N_{root}$

**Output:** The resulting community  $C$

// Fine-tuning Phase

- 1  $\{\hat{c}(v) | v \in \mathcal{V}\} \leftarrow$  predict corenesses using  $N_{root}$  via Algorithm 1;
- 2  $V_{pos}, V_{neg}, V_{temp} \leftarrow$  sample by rules 1 and 2;
- 3 **for**  $epoch = 1$  to  $\lambda$  **do**
- 4    $\mathcal{L} \leftarrow 0$ ;  $\{\mathbf{h}^{(l)}(v) | v \in \mathcal{V}\} \leftarrow$  infer with  $\mathcal{M}_{AT}$  via Algorithm 2;
- 5    $\mathcal{L} \leftarrow$  compute the loss function based on  $V_{pos}, V_{neg}, V_{temp}$ ;
- 6    $\theta(\mathcal{M}_{ADP}) \leftarrow \theta(\mathcal{M}_{ADP}) - \varphi \cdot \nabla_{\theta} \mathcal{L}$ ;

// Searching Phase

- 7  $C \leftarrow V_Q$ ;  $\{\mathbf{h}^{(l)}(v) | v \in \mathcal{V}\} \leftarrow$  infer with  $\mathcal{M}_{AT}$  via Algorithm 2;
- 8  $V_r \leftarrow r$ -hop neighbors of  $V_Q$ ;
- 9  $\tau \leftarrow \frac{\sum_{v \in V_r} dis(v, V_Q)}{|V_r - V_Q|}$ ;  $min\_dis \leftarrow 0$ ;
- 10 **while**  $min\_dis < \tau$  **do**
- 11    $C \leftarrow C \cup \arg \min_{v \in N_{T_Q}(C) \cap V_r} dis(v, V_Q)$ ;
- 12    $min\_dis \leftarrow \min_{v \in N_{T_Q}(C) \cap V_r} dis(v, V_Q)$ ;
- 13    $\tau \leftarrow \cos(\frac{\pi}{2} \cdot \frac{|C|}{|V_r|}) \cdot \tau + (1 - \cos(\frac{\pi}{2} \cdot \frac{|C|}{|V_r|})) \cdot \frac{\sum_{v' \in C} dis(v', V_Q)}{|C|}$ ;
- 14 **return**  $C$ ;

**5.3 Searching Phase**

The searching phase identifies the final community for TICS. COCO progressively expands the result set by adding vertices whose embeddings are similar to those of query vertices. We introduce a novel adaptive search algorithm that dynamically adjusts the termination threshold.

We achieve this dynamic adjustment through an adaptive update function (Eq.(17)). Based on small-world theory [16, 25], community members exhibit strong locality, so we first extract an  $r$ -hop subgraph around query vertices, denoted as  $V_r$ . Initially, the threshold  $\tau$  is set to the average embedding distance between vertices in the  $r$ -hop subgraph  $G_r = (V_r, E_r)$  and query vertices  $V_Q$ :  $\tau = \frac{\sum_{v \in V_r} dis(v, V_Q)}{|V_r - V_Q|}$ . During each iteration, COCO selects the vertex with the minimum distance and adds it to the result set  $C$ . For the  $n$ -th iteration, when a new vertex is added to  $C$ , the threshold  $\tau$  is updated according to the following function:

$$\tau_n = \rho_n \cdot \tau_{n-1} + (1 - \rho_n) \cdot \frac{\sum_{v \in C} dis(v, V_Q)}{|C|}, \quad \rho_n = \cos\left(\frac{\pi}{2} \cdot \frac{|C|}{|V_r|}\right) \quad (17)$$

Here,  $\rho_n$  smooths the update by combining the previous threshold with the current average distance in  $C$ . This prevents abrupt changes as  $C$  grows and allows the threshold to adapt to the current community instead of using a fixed cutoff. The search terminates when the minimum distance among the remaining candidates exceeds  $\tau_n$ . The following lemma characterizes how the smoothing factor changes with  $|C|$  and the limiting threshold used by this stopping rule.

**LEMMA 5.1.** *The smoothing factor  $\rho_n = \cos\left(\frac{\pi}{2} \cdot \frac{|C|}{|V_r|}\right)$  decreases monotonically with community size  $|C|$  and satisfies  $\lim_{|C| \rightarrow |V_r|} \rho_n = 0$ , guaranteeing that  $\tau$  converges to  $\frac{\sum_{v \in C} dis(v, V_Q)}{|C|}$ .*

**PROOF.** As  $|C|$  increases from  $|V_Q|$  to  $|V_r|$ ,  $\frac{\pi}{2} \cdot \frac{|C|}{|V_r|}$  grows from  $\frac{\pi}{2} \cdot \frac{|V_Q|}{|V_r|}$  to  $\frac{\pi}{2}$ . Within this interval, the cosine function is strictly

**Table 2: Statistics of networks**

| Dataset       | Abbr. | $ \mathcal{V} $ | $ \mathcal{E} $ | $t_{max}$ | GCC    | Null                 | Ratio |
|---------------|-------|-----------------|-----------------|-----------|--------|----------------------|-------|
| CollegeMsg    | CM    | 1.8K            | 59.8K           | 193       | 0.0637 | 0.0146               | 4.4   |
| MathOverflow  | MO    | 24.8K           | 506.5K          | 2351      | 0.1512 | 0.0220               | 6.8   |
| Superuser     | SU    | 194.1K          | 1.4M            | 2774      | 0.0219 | 0.0061               | 3.6   |
| WikiTalk      | WT    | 1.1M            | 7.8M            | 2321      | 0.0209 | $7.8 \times 10^{-4}$ | 27.0  |
| Youtube       | YT    | 3.2M            | 9.3M            | 226       | 0.0025 | $1.3 \times 10^{-4}$ | 19.5  |
| Flickr        | FL    | 2.3M            | 33.1M           | 394       | 0.0475 | 0.0127               | 3.8   |
| StackOverflow | SO    | 2.6M            | 63.5M           | 2774      | 0.0169 | 0.0055               | 3.1   |
| Wikipedia     | WP    | 6.9M            | 129.9M          | 298       | 0.0792 | 0.0112               | 7.1   |

decreasing, which means  $\rho_n$  strictly decreases and approaches 0. In the update equation, when  $\rho_n$  approaches 0, the term  $\rho_n \cdot \tau_{n-1}$  becomes negligible while  $(1 - \rho_n)$  tends toward 1. Consequently,  $\tau_n$  becomes increasingly determined by the current average distance  $\frac{\sum_{v \in C} dis(v, V_Q)}{|C|}$ . Since vertices are added in order of increasing distance, this average distance is non-decreasing and eventually stabilizes, causing  $\tau$  to converge to this stabilized average distance.  $\square$

The cosine form is chosen for its slope. Early in the search, it changes slowly and keeps the threshold stable for the small seed set. Later, it decreases faster and gives more weight to the distances observed in  $C$ , which curbs excessive expansion.

Overall, the search process is presented in lines 7-14 of Algorithm 5. First, the result set is initialized with query vertices. Embeddings of all vertices are obtained through the inference of AT-GNN (line 7). Next, the  $r$ -hop subgraph of is extracted (line 8). The search threshold is then set to the average distance within the subgraph (line 9). The algorithm adds the neighboring vertex with the smallest distance into the result set (line 11). After that, the minimal distance is updated (line 12), and the threshold is adjusted accordingly (line 13). Once the minimal distance exceeds the threshold, the search process stops.

**5.4 Time Complexity Analysis**

**Pre-training Phase** The time complexity contains two components. First, the HM-Index tree construction involves building a tree structure with depth  $h = \log_p \frac{t_{max}}{x}$ , where  $t_{max}$  is the maximal timestamp,  $x$  is the leaf interval length, and  $p$  is the branching factor. The core decomposition process requires  $O(|\mathcal{V}| + |\mathcal{E}|)$  time in total across all timestamps. The HM-Index structure contains  $O(\frac{t_{max}}{x})$  nodes. For each node, the training process requires  $O(|\mathcal{V}| \cdot \log \frac{t_{max}}{x})$  time per epoch. This leads to an overall time complexity of  $O(|\mathcal{E}| + \lambda \cdot \frac{t_{max}}{x} \cdot |\mathcal{V}| \cdot \log \frac{t_{max}}{x})$ . Second, the AT-GNN processes  $K$  GNN layers with feature dimension  $d$ . Each layer requires  $O(|\mathcal{V}| + |\mathcal{E}|)$  time. Training on  $Q$  queries, this results in a time complexity of  $O(\lambda \cdot Q \cdot K \cdot (|\mathcal{V}| + |\mathcal{E}|))$ . Overall, the total time complexity is  $O(\lambda \cdot \frac{t_{max}}{x} \cdot |\mathcal{V}| \cdot \log \frac{t_{max}}{x} + \lambda \cdot Q \cdot K \cdot (|\mathcal{V}| + |\mathcal{E}|))$ .

**Fine-tuning Phase.** The computational cost in this phase is dominated by updating the parameters in the adapters, while the encoder parameters remain frozen. Consequently, fine-tuning the adapters requires  $O(\lambda \cdot Q \cdot K \cdot |\mathcal{V}|)$  time.

**Searching Phase.** The process begins with predicting vertex corenesses in the  $r$ -hop subgraph using the HM-Index, which requires  $O(|V_r| \cdot \log \frac{t_{max}}{x})$  time. Next, vertex embeddings are generated by the AT-GNN and subsequently sorted, taking  $O(K \cdot (|V_r| + |E_r|) + |V_r| \cdot \log |V_r|)$  time. Finally, the search process completes in  $O(|V_r|)$  iterations. Therefore, the overall time complexity is  $O(K \cdot (|V_r| + |E_r|) + |V_r| \cdot (\log |V_r| + \log \frac{t_{max}}{x}))$ .

**Table 3: Community quality comparison for different methods**

| Datasets                | CM            |               |               | MO            |               |               | SU            |               |               | WT            |               |               | YT            |               |               | FL            |               |               | SO            |               |               | WP            |               |               | Avg +/- |        |         |
|-------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------|--------|---------|
| Metrics                 | TD            | TC            | TM            | TD            | TC            | TM            | TD            | TC            | TM            | TD            | TC            | TM            | TD            | TC            | TM            | TD            | TC            | TM            | TD            | TC            | TM            | TD            | TC            | TM            | TD      | TC     | TM      |
| COCO                    | <b>0.0064</b> | <b>0.5625</b> | <b>0.0375</b> | <b>0.0018</b> | <b>0.5715</b> | <b>0.0280</b> | <b>0.0011</b> | <b>0.5872</b> | <b>0.0114</b> | <b>0.0042</b> | <b>0.6441</b> | <b>0.0085</b> | <b>0.0065</b> | <b>0.6862</b> | <b>0.0045</b> | <b>0.0049</b> | <b>0.7624</b> | <b>0.0020</b> | <b>0.0097</b> | <b>0.5225</b> | <b>0.0042</b> | <b>0.0123</b> | <b>0.5947</b> | <b>0.0011</b> | -       | -      | -       |
| Temporal <i>k</i> -core | 0.0044        | 0.6771        | 0.0219        | 0.0006        | 0.7292        | 0.0090        | 0.0001        | 0.6762        | 0.0010        | 0.0016        | 0.8304        | 0.0028        | 0.0008        | 0.7443        | 5.1e-4        | 0.0001        | 0.8263        | 6.3e-5        | 0.0031        | 0.7231        | 0.0012        | 0.0051        | 0.8308        | 4.2e-4        | -69.8%  | +23.4% | -73.4%  |
| QTCS                    | 0.0048        | 0.6332        | 0.0255        | 0.0011        | 0.6572        | 0.0161        | 0.0006        | 0.6580        | 0.0058        | 0.0022        | 0.7408        | 0.0041        | 0.0028        | 0.8676        | 0.0016        | 0.0033        | 0.8695        | 0.0011        | 0.0045        | 0.7786        | 0.0017        | OOM           | OOM           | OOM           | -43.2%  | +20.6% | -48.6%  |
| TransZero               | 0.0026        | 0.8433        | 0.0051        | 0.0002        | 0.8999        | 0.0021        | 9.32e-5       | 0.9293        | 6.9e-4        | 0.0001        | 0.8838        | 1.8e-4        | 0.0004        | 0.9650        | 1.9e-4        | 0.0003        | 0.9865        | 4.5e-5        | 7.02e-5       | 0.9780        | 2.4e-5        | OOM           | OOM           | OOM           | -89.1%  | +51.4% | -94.8%  |
| CS-Former               | 0.0007        | 0.9037        | -0.0085       | 0.0003        | 0.9490        | 4.9e-4        | 8.05e-6       | 0.9660        | -5.8e-4       | 0.0002        | 0.9388        | 2.1e-4        | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | -94.7%  | +59.2% | -105.9% |
| IACS                    | 0.0030        | 0.9500        | -0.0036       | 0.0002        | 0.8340        | 0.0019        | 0.0001        | 0.9895        | -6.8e-4       | 0.0004        | 0.6850        | 4.5e-4        | 1.98e-7       | 0.8785        | 7.7e-7        | 8.04e-8       | 0.9872        | 2.2e-7        | 1.03e-6       | 0.8942        | 9.3e-7        | OOM           | OOM           | OOM           | -90.7%  | +45.5% | -100.5% |
| CS-TGN                  | 0.0024        | 0.9059        | 0.0014        | 5.12e-5       | 0.7008        | 0.0017        | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | -78.6%  | +41.8% | -95.0%  |
| UTCS                    | 0.0019        | 0.6876        | 0.0118        | 0.0016        | 0.6678        | 0.0209        | 0.0003        | 0.6481        | 0.0030        | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | OOM           | -51.9%  | +16.5% | -56.0%  |

\* A community is considered to be of higher quality when it exhibits a **higher TD**, a **lower TC**, and a **higher TM**.

## 6 Performance Evaluation

This section evaluates the performance of COCO. Our experiments utilize eight real-world temporal networks, whose statistical properties are summarized in Table 2. All experiments were conducted on an Ubuntu server, equipped with an Intel i9-10900K CPU, 128GB of memory, and an Nvidia RTX 3090 GPU. More experimental results can be found in the *technical report* [1]. The source code and implementation instructions are available at <https://anonymous.4open.science/r/COCO-F183>.

**Competitors.** We compare COCO with seven competitors.

- *QTCS* [21]. A non-learning method that uses time-constrained personalized pagerank to find a  $\beta$ -temporal proximity core containing the query vertex.
- *Temporal  $k$ -core* [43]. A non-learning method that identifies a  $k$ -core in the projected graph of the given time interval. We employ an indexing-based method [44] to speed up its search process.
- *TransZero* [35]. An unsupervised learning framework for community search using graph transformers.
- *CSFormer* [39]. An unsupervised learning-based approach for static graphs that adds vertices whose predicted corenesses are higher than those of the query vertices into the community.
- *IACS* [5]. An inductive learning-based method for community search across different static graphs.
- *CS-TGN* [10]. A supervised learning-based method designed for community search over temporal graphs.
- *UTCS* [45]. An unsupervised method for temporal graphs that pre-trains on temporal dynamics and subgraph knowledge to identify communities using a proximity-based scoring mechanism.

**Metrics.** Given the absence of ground-truth datasets for temporal graphs, we follow QTCS [21] and use Temporal Density (TD) and Temporal Conductance (TC). We additionally use Temporal Modularity (TM) for community search [15], which extends classic modularity [27] by treating interaction counts in the query interval as edge weights. For a query interval  $[t_s, t_e]$ , let  $\mathcal{E}_Q = \mathcal{E}_{[t_s, t_e]}$ . Given a retrieved community  $C$ , the three metrics are defined as follows.

$$TD(C) = \frac{2 \cdot |\{(v, u, t) \in \mathcal{E}_{[t_s, t_e]} \mid v, u \in C\}|}{|C| \cdot (|C| - 1) \cdot (t_e - t_s + 1)} \quad (18)$$

$$TC(C) = \frac{|\{(v, u, t) \in \mathcal{E}_{[t_s, t_e]} \mid v \in C, u \notin C\}|}{\min \left\{ \sum_{v \in C} |\{(v, u, t) \in \mathcal{E}_{[t_s, t_e]}\}|, \sum_{v \notin C} |\{(v, u, t) \in \mathcal{E}_{[t_s, t_e]}\}| \right\}} \quad (19)$$

$$TM(C) = \frac{|\{(v, u, t) \in \mathcal{E}_Q \mid v, u \in C\}|}{|\mathcal{E}_Q|} - \left( \frac{\sum_{v \in C} |\{(v, u, t) \in \mathcal{E}_Q\}|^2}{2|\mathcal{E}_Q|} \right) \quad (20)$$

**TD** measures the internal cohesion of  $C$  as the fraction of possible internal temporal edges that occur during the query interval. **TC** measures the external separation of  $C$  by comparing the number of temporal edges crossing its boundary with the smaller incident temporal-edge volume. **TM** compares the fraction of interactions within  $C$  with its expectation under a degree-preserving null model. Accordingly, a high-quality community has a high TD, a low TC, and a high TM.

**Settings.** We set the HM-Index partition coefficient  $p = 4$  and leaf threshold  $x = 40$  to balance construction efficiency and query speed. The learning rate is 0.001. AT-GNN samples 200 random vertices and 200 time intervals for training, while COCO runs for 150 epochs with early stopping. We apply 10 fine-tuning steps. During search, we extract a 5-hop subgraph per query vertex and set  $\gamma = 1$  in Eq.(17). All baselines follow their published defaults. QTCS uses its standard settings, temporal  $k$ -core sets  $k$  to the query vertex coreness, and label-dependent models (IACS [5], CS-TGN [10]) generate pseudo-labels via TransZero [35]. For Exp-1, each query interval covers 10%–20% of the graph lifespan. Each experiment averages results over 100 random single-vertex queries.

**Community Tendency of the Datasets.** For each query interval, we compute the global clustering coefficient (GCC). We form a temporal null model by shuffling timestamps among temporal edge events while retaining their endpoints, preserving full-lifespan interactions but removing temporal localization. Table 2 reports averages over 100 query intervals. Across the eight datasets, GCC ranges from 0.0025 to 0.1512 and is 3.1–27.0 times the null value. These values indicate temporal clustering beyond the null expectation and meaningful community structure.

### 6.1 Effectiveness Evaluation

First, we evaluate the overall effectiveness of COCO.

**Exp-1: Community Quality Comparison.** We evaluate community quality for all methods using TD, TC, and TM. Table 3 reports the results across all datasets. We can observe that COCO demonstrates consistent superiority, outperforming all competitors in cohesion (TD), separation (TC), and modularity (TM). For instance, compared with the non-learning method QTCS, COCO has +43.22% TD improvement, +20.59% TC improvement, and +48.58% TM improvement. Compared with the learning method TransZero, COCO has +89.14% TD improvement, +51.40% TC improvement, and +94.78% TM improvement. COCO's dominance stems from the coupling of its two components: (1) HM-Index-derived coreness ensures cohesive community structures and (2) AT-GNN captures

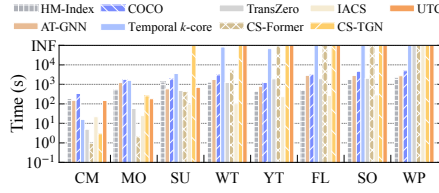


Figure 5: Training/Construction time

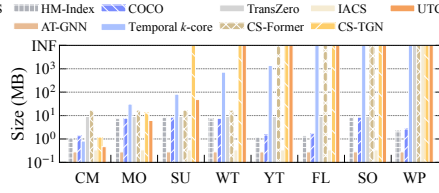


Figure 6: Model/Index Size

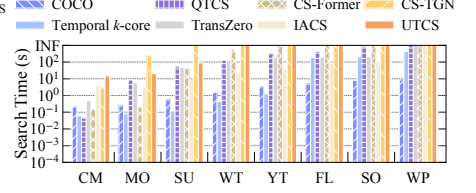


Figure 7: Search time

fine-grained evolutionary patterns. In contrast, methods such as temporal  $k$ -core and QTCS over-constrain communities via rigid  $k$ -core structures, excluding well-connected vertices. Additionally, learning-based approaches including TransZero, IACS, and CS-TGN fail to explicitly model structural cohesion, degrading performance.

## 6.2 Efficiency Evaluation

We evaluate the efficiency of COCO against other methods. In the experiments, we mark the algorithm as INF if it is out of memory or its running time exceeds  $10^5$  seconds.

**Exp-3: Offline Cost Comparison.** We compare the offline cost of different methods, including training or index construction time and model or index size. The results are shown in Figures 5 and 6. **Training / Construction Time** (Figure 5). For temporal  $k$ -core, we report index construction time. For learning-based methods, including COCO, TransZero, CS-Former, IACS, and CS-TGN, we report training time. COCO achieves training efficiency comparable to other learning-based methods, showing good scalability. In contrast, temporal  $k$ -core requires much longer construction time, especially on large datasets, because it must compute vertex core-ness over all time intervals. **Model / Index Size** (Figure 6). COCO is also compact. Its AT-GNN is smaller than the models used by other learning-based methods. Although HM-Index adds storage for the tree structure and node-level MLPs, the overall memory cost remains low because these MLPs are lightweight. By contrast, temporal  $k$ -core uses much more memory, since it stores core-ness values for all vertices over all time intervals, and its space cost grows quickly with graph size.

**Exp-4: Online Search Time Comparison.** COCO’s search time includes the fine-tuning time. The results are shown in Figure 7. Compared with non-learning methods, COCO demonstrates substantially better online efficiency. Specifically, COCO is approximately two orders of magnitude faster than QTCS. Compared with temporal  $k$ -core, COCO is slower on small temporal graphs (i.e., CM, MO, SU, WT, and YT) but faster on large temporal graphs (i.e., FL, SO, and WP). The reason is that, on small temporal graphs, the temporal  $k$ -core index can be successfully constructed, allowing answers to be retrieved directly. However, on large temporal graphs, the temporal  $k$ -core index cannot be built due to its high construction and storage cost, and thus temporal  $k$ -core has to perform online decomposition, which significantly increases query time.

**Exp-5: Effect of  $|T_Q|$  on Search Time.** We investigate the impact of the query time interval length, denoted as  $|T_Q|$ , on search efficiency by evaluating query intervals ranging from 20% to 80% of  $t_{\max}$ . Figure 8 presents the corresponding results on the FL, SO, and WP datasets. From these experiments, we obtain two key findings. First, all methods require longer search times as the query time interval increases, since broader temporal ranges involve processing

a larger number of vertices and edges. Second, COCO consistently outperforms the baseline methods by at least one order of magnitude, primarily due to its efficient fine-tuning mechanism.

**Exp-6: Scalability.** Figure 9 shows the scalability evaluation results of all methods on the FL, SO, and WP datasets. In this experiment, we randomly sampled vertices from the original graph to create graphs of varying sizes. As expected, the performance of all methods degrades as the graph size increases, since larger graphs usually contain larger communities. However, COCO maintains at least a  $10\times$  speed advantage over all baselines. This demonstrates COCO’s scalability for handling graphs of different scales.

## 6.3 Parameters Analysis

We analyze COCO’s parameter sensitivity by varying several key parameters. For the HM-Index, we adjust two parameters: the partition coefficient  $p$  and the leaf node threshold  $x$ . For AT-GNN, we consider two parameters:  $\delta$ , which adjusts the loss weights in Eq. (16), and  $\gamma$ , which controls the search process in Eq. (17).

**Exp-7: Effect of Partition Coefficient  $p$ .** The effect of the partition coefficient  $p$  in the HM-Index is shown in Figures 10 and 11, with the TM results reported in Figure 23(a) in Appendix D. As  $p$  increases, community quality improves consistently on all datasets, with higher TD, lower TC, and generally higher TM. This is mainly because a larger  $p$  creates more branches in the hierarchical tree and produces finer time partitions. As a result, the regression model at each node can better capture temporal and structural patterns within each sub-interval, improving core-ness prediction. Overall, HM-Index is robust to  $p$ , since the changes in community quality are limited. We set  $p = 4$  by default because it balances prediction reliability and index complexity.

**Exp-8: Effect of Leaf Node Threshold  $x$ .** The effect of the leaf node threshold  $x$  on community quality is shown in Figures 12 and 13, with the TM results reported in Figure 23(b) in Appendix D. Community quality slightly declines as  $x$  increases from 20 to 60 across all datasets, as indicated by the changes in TD, TC, and TM. A smaller  $x$  yields finer temporal partitions in the HM-Index leaf nodes, allowing regression models to better capture short-term dynamics and structural variations in vertex core-ness. In contrast, larger  $x$  leads to coarser intervals that may miss subtle temporal patterns, causing a minor accuracy drop. Nevertheless, community quality remains good. We choose  $x = 40$  as the default to balance prediction performance and indexing efficiency.

**Exp-9: Effect of  $\delta$ .** We evaluate the impact of  $\delta$ , which balances cohesive loss and temporal link loss in Eq. (16). Figures 14 and 15 present the TD and TC results, while Figure 23(c) in Appendix D reports TM. Community quality generally improves as  $\delta$  approaches 0.5 and degrades at higher values, with TM following the same

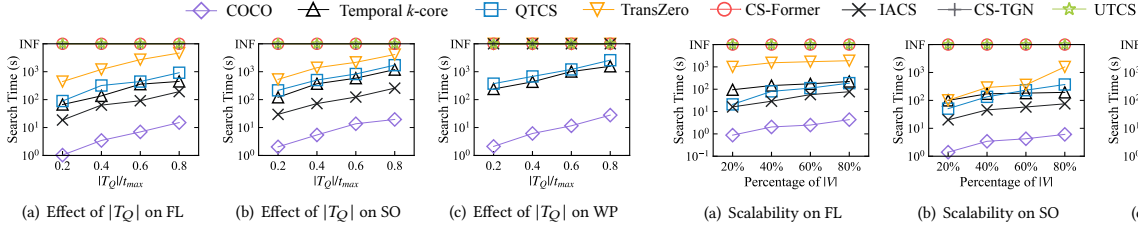
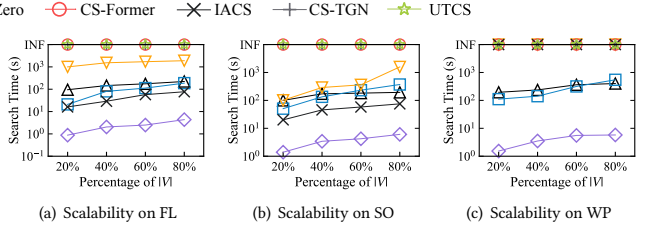
Figure 8: Effect of  $|T_Q|$ 

Figure 9: Scalability

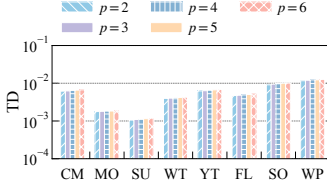
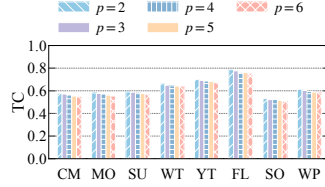
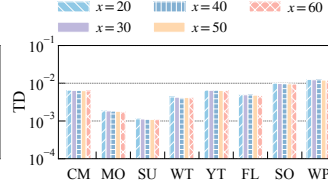
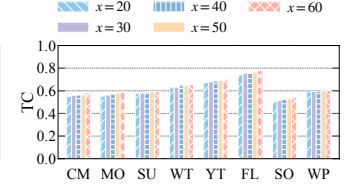
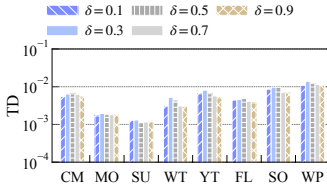
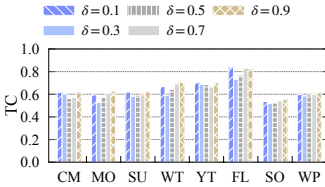
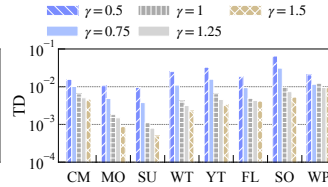
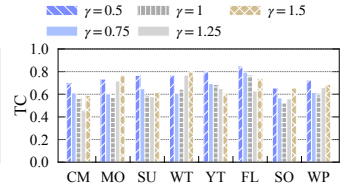
Figure 10: Effect of  $p$  on TDFigure 11: Effect of  $p$  on TCFigure 12: Effect of  $x$  on TDFigure 13: Effect of  $x$  on TCFigure 14: Effect of  $\delta$  on TDFigure 15: Effect of  $\delta$  on TCFigure 16: Effect of  $\gamma$  on TDFigure 17: Effect of  $\gamma$  on TC

Table 4: Component Ablation

| Methods                                                 | CM     |        |        | MO     |        |        | SU     |        |        | WT     |        |        | YT     |        |        | FL     |        |        | SO     |        |        | WP     |        |        | Avg. +/- |        |        |
|---------------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|----------|--------|--------|
| Metrics                                                 | TD     | TC     | TM     | TD     | TC     | TM     | TD     | TC     | TM     | TD     | TC     | TM     | TD     | TC     | TM     | TD     | TC     | TM     | TD     | TC     | TM     | TD     | TC     | TM     | TD       | TC     | TM     |
| COCO (full)                                             | 0.0064 | 0.5625 | 0.0375 | 0.0018 | 0.5715 | 0.0280 | 0.0011 | 0.5872 | 0.0114 | 0.0042 | 0.6441 | 0.0085 | 0.0065 | 0.6862 | 0.0045 | 0.0049 | 0.7624 | 0.0020 | 0.0097 | 0.5225 | 0.0042 | 0.0123 | 0.5947 | 0.0011 | -        | -      | -      |
| w/o HM-Index (use single MLP)                           | 0.0034 | 0.6982 | 0.0246 | 0.0013 | 0.8010 | 0.0172 | 0.0005 | 0.8240 | 0.0074 | 0.0031 | 0.9407 | 0.0053 | 0.0042 | 0.8292 | 0.0043 | 0.0020 | 0.9095 | 0.0014 | 0.0077 | 0.7652 | 0.0032 | 0.0066 | 0.7325 | 0.0007 | -39.8%   | +32.5% | -35.8% |
| w/o AT-GNN (use GAT)                                    | 0.0051 | 0.5935 | 0.0338 | 0.0015 | 0.7685 | 0.0234 | 0.0010 | 0.5953 | 0.0105 | 0.0038 | 0.7887 | 0.0072 | 0.0057 | 0.7803 | 0.0042 | 0.0046 | 0.8742 | 0.0023 | 0.0093 | 0.5777 | 0.0041 | 0.0097 | 0.7629 | 0.0009 | -12.2%   | +16.2% | -14.0% |
| w/o Adapter (use linear layer)                          | 0.0057 | 0.6208 | 0.0364 | 0.0016 | 0.6522 | 0.0253 | 0.0011 | 0.6037 | 0.0112 | 0.0038 | 0.6491 | 0.0083 | 0.0065 | 0.6943 | 0.0054 | 0.0048 | 0.8639 | 0.0021 | 0.0084 | 0.5237 | 0.0044 | 0.0110 | 0.6355 | 0.0009 | -7.5%    | +6.2%  | -6.7%  |
| w/o DFT (use raw temporal feature)                      | 0.0059 | 0.5989 | 0.0327 | 0.0017 | 0.6064 | 0.0226 | 0.0010 | 0.6302 | 0.0103 | 0.0039 | 0.7080 | 0.0075 | 0.0061 | 0.7508 | 0.0041 | 0.0046 | 0.8347 | 0.0024 | 0.0090 | 0.5711 | 0.0043 | 0.0115 | 0.6459 | 0.0008 | -6.51%   | +8.33% | -8.4%  |
| w/o $\mathcal{L}_{Coh}$ (use $\mathcal{L}_{Temp}$ only) | 0.0041 | 0.5856 | 0.0325 | 0.0017 | 0.5959 | 0.0245 | 0.0009 | 0.7158 | 0.0108 | 0.0036 | 0.7176 | 0.0073 | 0.0041 | 0.7662 | 0.0046 | 0.0041 | 0.8366 | 0.0022 | 0.0066 | 0.6392 | 0.0045 | 0.0113 | 0.5952 | 0.0008 | -20.8%   | +10.7% | -16.5% |
| w/o $\mathcal{L}_{Temp}$ (use $\mathcal{L}_{Coh}$ only) | 0.0056 | 0.6655 | 0.0332 | 0.0017 | 0.6958 | 0.0238 | 0.0009 | 0.6470 | 0.0094 | 0.0035 | 0.7731 | 0.0076 | 0.0059 | 0.6938 | 0.0043 | 0.0039 | 0.9249 | 0.0025 | 0.0082 | 0.5892 | 0.0042 | 0.0118 | 0.7635 | 0.0008 | -12.2%   | +16.7% | -16.2% |
| w/o Coreness Feature (use Node2Vec)                     | 0.0051 | 0.6101 | 0.0329 | 0.0014 | 0.6565 | 0.0231 | 0.0010 | 0.6401 | 0.0106 | 0.0028 | 0.6955 | 0.0071 | 0.0060 | 0.9055 | 0.0047 | 0.0045 | 0.9482 | 0.0026 | 0.0077 | 0.5794 | 0.0046 | 0.0109 | 0.6431 | 0.0008 | -16.6%   | +14.5% | -16.2% |

overall pattern as TD and TC. This is because excessive temporal loss weighting ( $\delta > 0.5$ ) undermines structural cohesion. The results establish  $\delta = 0.5$  as the optimal trade-off, simultaneously maximizing community cohesiveness and temporal consistency across most datasets.

**Exp-10: Effect of  $\gamma$ .** We study the effect of  $\gamma$ , which controls the threshold decay rate in the search function (Eq. (17)), by varying it from 0.5 to 1.5. Figures 16 and 17 report TD and TC, while Figure 23(d) in Appendix D reports TM. Increasing  $\gamma$  generally reduces TD and TM by producing larger communities, while TC shows dataset-dependent fluctuations. Notably, COCO maintains robust performance with the default  $\gamma = 1$ , demonstrating its stability across parameter choices.

## 6.4 Ablation Study

**Exp-11: Coreness Prediction Accuracy of HM-Index.** To justify the hierarchical design of HM-Index, we compare it with a global MLP trained on random intervals from the full graph lifespan and a nearest-neighbor (NN) method. NN caches exact coreness for the same hierarchical anchor intervals used by HM-Index ( $p = 4$  and  $x = 40$ ). For each  $T_Q$ , NN selects the cached anchor with

the largest interval IoU and returns the exact coreness stored for that anchor. We measure accuracy by  $1 - \text{MAPE}$ , where  $\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{c(v)_i - \hat{c}(v)_i}{c(v)_i} \right|$ . As shown in Figure 18, HM-Index consistently achieves over 90% accuracy, while both baselines perform much worse. The global MLP lacks enough capacity to handle the topological changes caused by query intervals of different lengths. The NN method is also inaccurate because temporal  $k$ -core structures are highly sensitive to edge changes. Even with large temporal overlap, missing a few critical edges can trigger cascading neighbor removals and break the core structure. HM-Index avoids these problems through its hierarchical design.

## Exp-12: Effect of Coreness Accuracy on Community Quality.

To evaluate how coreness prediction accuracy affects the final community quality, we compare the downstream TD and TC obtained using coreness values from HM-Index, NN, Global MLP, and exact coreness computation. As shown in Figures 19 and 20, NN and Global MLP lead to poor community quality across all datasets, yielding much lower TD and much higher TC. This is because their inaccurate coreness predictions substantially distort the AT-GNN feature space. In contrast, HM-Index produces results that

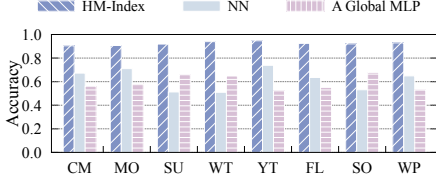


Figure 18: Coreness prediction accuracy

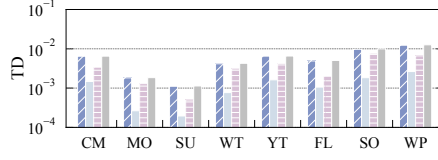


Figure 19: Effect of coreness prediction on TD

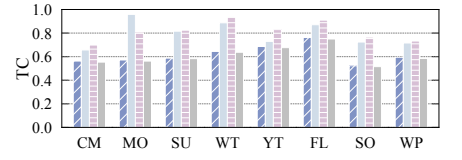


Figure 20: Effect of coreness prediction on TC

are nearly identical to those obtained with exact coreness. These results suggest two conclusions. First, accurate coreness prediction is necessary for identifying cohesive communities. Second, the prediction accuracy of HM-Index is high enough to preserve strong downstream community quality, without incurring the heavy cost of online exact core decomposition. On MO, NN has higher average coreness accuracy than Global MLP but yields worse TD and TC because the two predictors produce different error distributions. Interval mismatch can produce a few large NN errors that strongly distort the structural features used by AT-GNN, whereas Global MLP produces smoother errors that AT-GNN’s learned attention can partly tolerate. Therefore, average coreness accuracy alone does not determine downstream community quality.

**Exp-13: Component Ablation.** We evaluate the main components of COCO using TD, TC, and TM, including the hierarchical model index, AT-GNN, the adapter and frequency modeling, the loss functions, and the input features. The results are shown in Table 4. (1) Replacing HM-Index with a single MLP decreases TD by 39.78% and TM by 35.75% on average. This confirms the importance of hierarchical coreness prediction. (2) Replacing AT-GNN with GAT decreases TD by 12.15%, increases TC by 16.17%, and decreases TM by 14.00%. (3) Replacing the adapter with a linear layer removes the query-context feature ( $s_Q$ ) and the bottleneck structure, causing consistent degradation across the three metrics. (4) Replacing DFT-based modeling with raw temporal features decreases TD by 6.51%, increases TC by 8.33%, and decreases TM by 8.41%, showing the value of frequency-domain features for capturing interaction patterns. (5) Removing  $\mathcal{L}_{Coh}$  and  $\mathcal{L}_{Temp}$  decreases TM by 16.50% and 16.25%, respectively. The two losses provide distinct structural and temporal supervision. (6) Replacing the structural coreness feature with Node2Vec also degrades all three metrics, indicating that coreness is a more suitable signal for cohesive community search. Overall, every proposed component makes a contribution to COCO.

## 6.5 Case Study

**Exp-14: Case Study.** We conduct a case study on a temporal co-authorship network from DBLP [4], where vertices represent authors and timestamped edges denote publication collaborations. In the case study, we set the query time interval from 2020 to 2025. Figure 21 compares the communities returned by COCO, temporal  $k$ -core, QTCS, UTCS, and a frequency-only variant in four panels. The temporal  $k$ -core and QTCS results are overlaid in Figure 21(b). The frequency-only variant repeatedly adds the frontier author with the highest average interaction frequency. We compute one force-directed layout and fix the resulting vertex positions across all four panels, so differences between panels reflect the returned communities rather than independent layouts. Unselected vertices and edges are shown in gray, the query author is marked in red, and

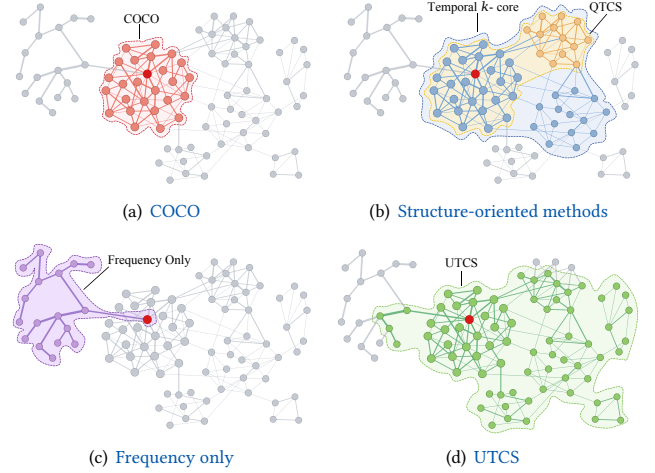


Figure 21: Case study on DBLP

edge thickness represents interaction frequency. Figure 21(b) shows that temporal  $k$ -core returns a broad community, while QTCS selects a smaller query-centered region within it. Nevertheless, both include collaboration groups outside the densely interacting center, since neither directly evaluates whether internal collaborations are occasional or repeated. The frequency-only result in Figure 21(c) follows several thick edges into a branching peripheral region, but its selected authors have weak mutual connectivity. UTCS in Figure 21(d) returns an even broader community containing multiple peripheral groups. In contrast, COCO in Figure 21(a) focuses on the dense central group, where the authors are mutually connected and many internal edges represent repeated collaborations. It therefore excludes both the peripheral authors retained by structure-oriented methods and the weakly connected branch selected using interaction frequency alone. These observations show that COCO identifies a more cohesive and active community than the baselines.

## 7 Conclusion

This paper addresses the problem of time-interval-based community search. We propose COCO, a cohesiveness-aware learning framework with three phases: pre-training, fine-tuning, and searching. Its coupling mechanism connects interval-specific coreness computation in HM-Index with query-specific temporal representation learning in AT-GNN. The predicted coreness serves as a structural prior, enabling COCO to identify communities with strong structural cohesiveness and frequent interactions. COCO avoids recomputing  $k$ -core or retraining the entire model for each query. COCO achieves both high-quality community finding and computational efficiency, as demonstrated through extensive experiments on real-world datasets.

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## A Discussion of Lemma 4.1

**Table 5: Error propagation through HM-Index**

| Dataset | $\max \beta_k$ | $\max \eta_k$ | $\bar{\epsilon}_h$ | $\bar{U}_h$ | Gap    |
|---------|----------------|---------------|--------------------|-------------|--------|
| CM      | 0.5683         | 0.2217        | 0.8912             | 1.2765      | 0.3859 |
| MO      | 0.6714         | 0.3448        | 0.8016             | 1.9911      | 1.1903 |
| SU      | 0.6821         | 0.3355        | 0.6438             | 2.0257      | 1.3824 |
| WT      | 0.7529         | 0.3942        | 0.9634             | 3.4826      | 2.5190 |
| YT      | 0.6836         | 0.3203        | 1.4071             | 2.2708      | 0.8635 |
| FL      | 0.6587         | 0.3820        | 1.9959             | 2.7762      | 0.7816 |
| SO      | 0.6722         | 0.3434        | 0.6697             | 2.1693      | 1.5008 |
| WP      | 0.6605         | 0.3081        | 2.1496             | 2.8714      | 0.7227 |

**Contraction assumption.** Lemma 4.1 treats  $\beta < 1$  as a verifiable property of the hierarchical refinement process rather than a first-principles guarantee. The parent prediction enters Eq. (2) through  $\mathbf{W}_0 \hat{\mathbf{c}}$ , while each child model uses interval-specific features to refine the inherited estimate. This residual design makes contraction plausible but does not guarantee it. The evaluation below estimates a separate nonnegative slope  $\beta_k$  at each depth from consecutive prediction errors. Table 5 reports  $\max_k \beta_k$  rather than one pooled slope. The values range from 0.568 to 0.752 across the eight datasets and remain below 1.

**Overfitting at deeper nodes.** Every HM-Index node is trained with the same number of vertex batches of size  $n_b$  and the same  $\lambda$  training epochs, and each sampled vertex is paired with a randomly selected subinterval. Thus, the number of sampled vertex-interval pairs does not decrease with depth. The dimension of  $\mathbf{s}_2$  is bounded by  $p$ , while the maximum dimension of  $\mathbf{s}_1$  decreases with  $|T_N|$ . The model capacity therefore does not increase and generally decreases with depth, which reduces the risk of overfitting. Although the local signal may become less informative at deeper levels, the additive error term  $\eta$  accounts for the unresolved component in residual refinement.

**Tightness of the bound.** The term  $\eta/(1 - \beta)$  is a conservative worst-case bound obtained by taking  $\eta = \sup_j \eta_j$ . It need not be attained by the actual error sequence. Retaining the approximation error at each level gives the tighter finite-depth form

$$\epsilon_h \leq \beta^h \epsilon_0 + \sum_{k=0}^{h-1} \beta^{h-1-k} \eta_k.$$

For each depth, the evaluation below sets  $\eta_k$  to the largest positive residual after fitting  $\beta_k$ . Table 5 reports that the largest  $\eta_k$  values range from 0.221 to 0.394. The mean final errors range from 0.643 to 2.149, while the corresponding uniform bounds range from 1.276 to 3.482. Their mean gaps range from 0.385 to 2.519, and every evaluated pair has a nonnegative gap. These results are consistent with the bound and show that it is conservative.

**Empirical Evaluation of Error Propagation.** For every vertex-interval pair  $i$  used in the evaluation, we record the absolute prediction error  $\epsilon_{i,k}$  after each visited depth  $k$ . At each depth, we fit  $\epsilon_{i,k} = \beta_k \epsilon_{i,k-1} + r_{i,k}$  with  $\beta_k \geq 0$  and set  $\eta_k = \max_i \max\{0, r_{i,k}\}$ . This choice makes the resulting recurrence valid for every evaluated pair. We then use  $\beta = \max_k \beta_k$  and  $\eta = \max_k \eta_k$  to compute the

uniform bound  $U_{i,h}$  in Lemma 4.1. Table 5 reports  $\max_k \beta_k$ ,  $\max_k \eta_k$ , the mean observed final error, the mean uniform bound, and their mean gap.

## B Proof of Lemma 4.2

**PROOF OF LEMMA 4.2.** Let  $\hat{a}_k = \frac{1}{\sqrt{T}} \sum_{t=0}^{T-1} a_t e^{-2\pi i k t / T}$  denote the  $k$ -th coefficient of the unitary DFT of  $\mathbf{a}$ , and similarly for  $\hat{b}_k$ . Since the DFT is linear,  $\hat{a}_k - \hat{b}_k$  is the  $k$ -th DFT coefficient of the difference sequence  $\mathbf{a} - \mathbf{b}$ .

By the reverse triangle inequality for complex modulus, we have  $|\hat{a}_k| - |\hat{b}_k| \leq |\hat{a}_k - \hat{b}_k|$  for each frequency index  $k$ . Squaring and summing over the first  $d$  components yields  $\|\phi_d(\mathbf{a}) - \phi_d(\mathbf{b})\|_2^2 = \sum_{k=0}^{d-1} (|\hat{a}_k| - |\hat{b}_k|)^2 \leq \sum_{k=0}^{d-1} |\hat{a}_k - \hat{b}_k|^2$ .

Since  $d \leq T$ , the partial sum is bounded by the full sum:  $\sum_{k=0}^{d-1} |\hat{a}_k - \hat{b}_k|^2 \leq \sum_{k=0}^{T-1} |\hat{a}_k - \hat{b}_k|^2$ . By Parseval's identity for the unitary DFT, this equals  $\|\mathbf{a} - \mathbf{b}\|_2^2 = \sum_{t=0}^{T-1} (a_t - b_t)^2$ . Because  $\mathbf{a}, \mathbf{b} \in \{0, 1\}^T$ , each term  $(a_t - b_t)^2 \in \{0, 1\}$ , and exactly  $m$  terms are nonzero by assumption. Therefore  $\|\mathbf{a} - \mathbf{b}\|_2^2 = m$ .

Combining the above, we obtain  $\|\phi_d(\mathbf{a}) - \phi_d(\mathbf{b})\|_2^2 \leq m$ , which gives  $\|\phi_d(\mathbf{a}) - \phi_d(\mathbf{b})\|_2 \leq \sqrt{m}$ .  $\square$

## C Proof of Lemma 4.3

**PROOF.** For the fixed query context and ReLU activation region in the lemma, let  $\mathbf{J}_\Phi$  be the Jacobian of the fusion module with respect to its projected vertex input, and let  $\mathbf{D}_\sigma$  be the diagonal derivative matrix of the ReLU. The chain rule gives  $\frac{\partial \text{ADP}_Q}{\partial \mathbf{h}} = \mathbf{D}_\sigma \mathbf{W}_{\text{up}} \mathbf{J}_\Phi \mathbf{W}_{\text{down}_1}$ . The rank of a matrix product cannot exceed the rank of any factor. Because the signal passes through both the adapter bottleneck and the attention-head representation, the Jacobian rank is at most  $\min\{\dim_{\text{ADP}}, \dim_{\text{head}}\} = \xi$ .

For exact realization, consider an affine correction  $\Delta_Q(\mathbf{h}) = \mathbf{A}_Q \mathbf{h} + \mathbf{b}_Q$  whose output remains componentwise nonnegative in the fixed region. Augment the input as  $\tilde{\mathbf{h}} = [\mathbf{h}^\top \ 1]^\top$ . If  $r = \text{rank}([\mathbf{A}_Q \ \mathbf{b}_Q]) \leq \xi$ , then  $[\mathbf{A}_Q \ \mathbf{b}_Q] = \mathbf{U} \mathbf{V}^\top$  through an  $r$ -dimensional space. Write  $\mathbf{V}^\top = [\mathbf{V}_h^\top \ \mathbf{v}_0]$ . Embed  $\mathbf{V}_h^\top$  and  $\mathbf{U}$  into  $\mathbf{W}_{\text{down}_1}$  and  $\mathbf{W}_{\text{up}}$ , respectively, with zero padding when needed. Since  $\mathbf{s}_Q$  is fixed and nonzero for a valid query interval, choose  $\mathbf{W}_{\text{down}_2}$  such that  $\mathbf{W}_{\text{down}_2} \mathbf{s}_Q = \mathbf{v}_0$ . Under the linear/identity fusion configuration, these branches produce  $\mathbf{V}_h^\top \mathbf{h} + \mathbf{v}_0$ . The pre-activation output is therefore  $\mathbf{U}(\mathbf{V}_h^\top \mathbf{h} + \mathbf{v}_0) = \mathbf{A}_Q \mathbf{h} + \mathbf{b}_Q$ . Its componentwise nonnegativity makes the ReLU an identity on this region, proving exact realization.  $\square$

## D More Experimental Results

**Exp-2: Progressive Graph Randomization.** We progressively rewire the temporal edges of all eight datasets from 0% to 100%. Each swap uses two edges with the same timestamp and exchanges their endpoints while rejecting self-loops and duplicate edges. This preserves every vertex's temporal degree at each timestamp while weakening higher-order structure. We rebuild COCO for every randomized graph and use the same queries at all randomization levels. Figure 22 shows consistent changes across the eight datasets. At full randomization, TD decreases by 52.38%–58.16%, TM decreases

by 77.33%–83.64%, and TC increases by 18.31%–37.80%. This degradation shows that the community structure identified by COCO depends on clustering in the input graphs.

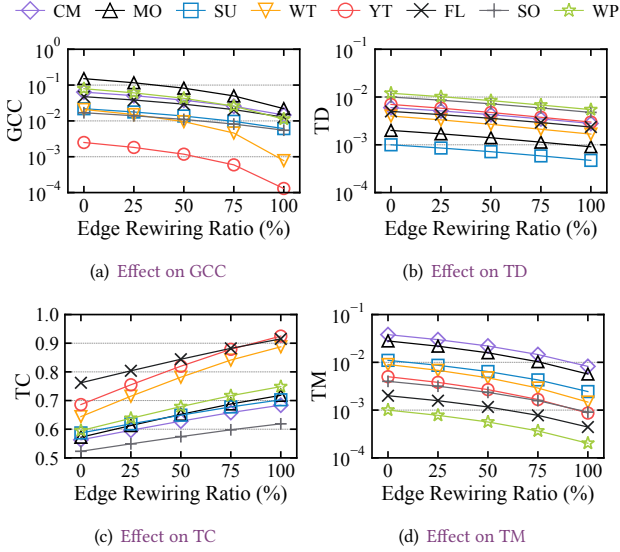


Figure 22: Effect of graph randomization

**Additional TM Results for the Parameter Analysis.** Figure 23 reports TM for the four parameters studied in Exp-7–Exp-10. The trends agree with the corresponding TD and TC results in the main paper.

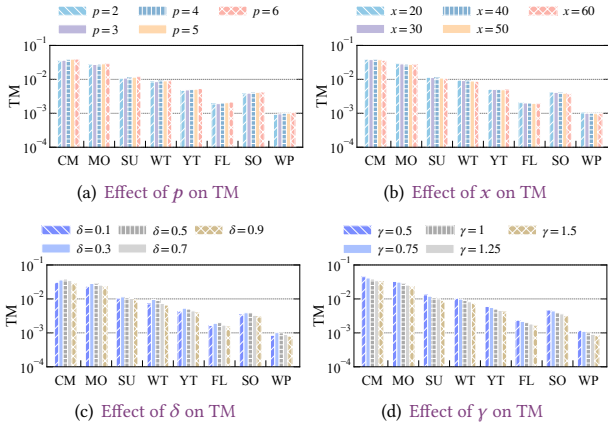


Figure 23: Effect of model parameters on temporal modularity

**Exp-15: Effect of  $|V_Q|$ .** Figure 24 reports TM as the number of query vertices varies. As  $|V_Q|$  increases, TM decreases slightly across all datasets. Larger query sets typically require more intermediate vertices to maintain structural connectivity and cohesiveness, expanding community size. The resulting communities have lower internal interaction density, which explains the small decrease in TM. Despite this, COCO maintains stable performance across query

scales, demonstrating its robustness in multi-vertex community search.

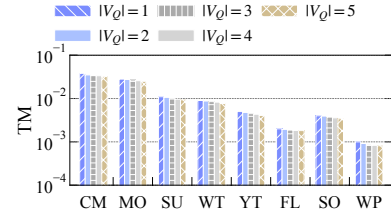


Figure 24: Effect of  $|V_Q|$  on temporal modularity

**Exp-16: Effect of # of Fine-tuning Steps.** We explore the influence of the number of fine-tuning steps on COCO. Specifically, we test 5, 10, 15, 20, and 25 steps. Figure 25 shows that TM increases as more steps are used. The largest gains occur between 10 and 15 steps, and the improvement becomes smaller thereafter. This suggests that COCO benefits from sufficient adaptation ( $\geq 15$  steps), while excessive fine-tuning yields limited additional improvement.

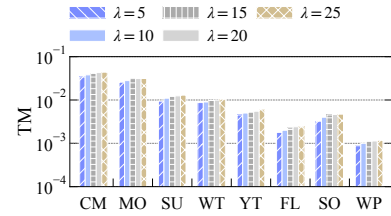


Figure 25: Effect of fine-tuning steps on temporal modularity

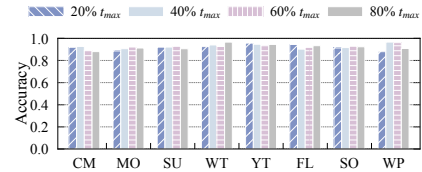


Figure 26: Effect of  $|T_Q|$  on prediction accuracy

**Exp-17: Effect of  $|T_Q|$  on Prediction Accuracy of HM-Index.** In this experiment, we evaluate the prediction accuracy of HM-Index by predicting vertex coreness across progressive time intervals (20%, 40%, 60%, and 80% of  $t_{max}$ ). Results shown in Figure 26 demonstrate consistent performance above 90% accuracy across all datasets and interval lengths. This robustness stems from HM-Index’s hierarchical architecture. COCO initiates its search from the root node, traverses through the HM-Index to identify nodes encompassing the query time interval. The models along the search path collectively refine coreness predictions.

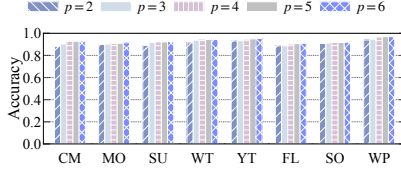
**Exp-18: Effect of  $p$  on Prediction Accuracy of HM-Index.** The effect of the partition coefficient  $p$  of HM-Index is shown in Figure 27. As observed in the figure, the prediction accuracy shows a consistent upward trend as  $p$  increases across all datasets. This improvement is primarily because a larger partition coefficient leads to more branches in the hierarchical tree, allowing for a finer-grained division of the time intervals. Such detailed partitioning enables

**Table 6: Community quality comparison for the worst 5% of queries**

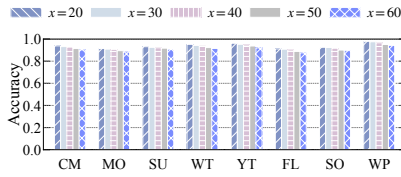
| Datasets        | CM                   |        |                      | MO                   |        |                      | SU                   |        |                       | WT                   |        |                      | YT                   |        |                      | FL                   |        |                      | SO                   |        |                      | WP     |        |                      | Avg +/- |        |         |
|-----------------|----------------------|--------|----------------------|----------------------|--------|----------------------|----------------------|--------|-----------------------|----------------------|--------|----------------------|----------------------|--------|----------------------|----------------------|--------|----------------------|----------------------|--------|----------------------|--------|--------|----------------------|---------|--------|---------|
| Metrics         | TD                   | TC     | TM                   | TD                   | TC     | TM                   | TD                   | TC     | TM                    | TD                   | TC     | TM                   | TD                   | TC     | TM                   | TD                   | TC     | TM                   | TD                   | TC     | TM                   | TD     | TC     | TM                   | TD      | TC     | TM      |
| COCO            | 0.0042               | 0.5837 | 0.0194               | 0.0013               | 0.5882 | 0.0146               | $6.2 \times 10^{-4}$ | 0.6105 | 0.0053                | 0.0024               | 0.6593 | 0.0042               | 0.0041               | 0.7118 | 0.0023               | 0.0032               | 0.7814 | $8.3 \times 10^{-4}$ | 0.0064               | 0.5457 | 0.0024               | 0.0072 | 0.6109 | $4.2 \times 10^{-4}$ | -       | -      | -       |
| Temporal k-core | 0.0023               | 0.7785 | 0.0094               | $2.7 \times 10^{-4}$ | 0.8173 | 0.0042               | $3.8 \times 10^{-5}$ | 0.7854 | $3.4 \times 10^{-4}$  | $6.9 \times 10^{-4}$ | 0.8956 | $8.9 \times 10^{-4}$ | $2.9 \times 10^{-4}$ | 0.8325 | $1.6 \times 10^{-4}$ | $3.5 \times 10^{-5}$ | 0.9034 | $1.8 \times 10^{-5}$ | 0.0012               | 0.8126 | $3.9 \times 10^{-4}$ | 0.0023 | 0.9015 | $1.5 \times 10^{-4}$ | -77.0%  | +33.3% | -79.5%  |
| QTCS            | 0.0032               | 0.7415 | 0.0124               | $5.8 \times 10^{-4}$ | 0.7553 | 0.0072               | $2.8 \times 10^{-4}$ | 0.7664 | 0.0023                | $9.8 \times 10^{-4}$ | 0.8315 | 0.0012               | 0.0012               | 0.9274 | $4.8 \times 10^{-4}$ | 0.0012               | 0.9283 | $3.2 \times 10^{-4}$ | 0.0023               | 0.8665 | $5.3 \times 10^{-4}$ | OOM    | OOM    | OOM                  | -54.8%  | +30.8% | -60.1%  |
| TransZero       | 0.0012               | 0.9174 | 0.0023               | $7.3 \times 10^{-5}$ | 0.9483 | $5.8 \times 10^{-4}$ | $2.8 \times 10^{-5}$ | 0.9643 | $1.9 \times 10^{-4}$  | $3.2 \times 10^{-5}$ | 0.9364 | $4.3 \times 10^{-5}$ | $1.3 \times 10^{-4}$ | 0.9843 | $4.2 \times 10^{-5}$ | $8.3 \times 10^{-5}$ | 0.9942 | $8.4 \times 10^{-6}$ | $1.9 \times 10^{-5}$ | 0.9903 | $4.2 \times 10^{-6}$ | OOM    | OOM    | OOM                  | -93.6%  | +52.3% | -97.2%  |
| CS-Former       | $2.4 \times 10^{-4}$ | 0.9543 | -0.0152              | $9.3 \times 10^{-5}$ | 0.9774 | $1.3 \times 10^{-4}$ | $2.3 \times 10^{-6}$ | 0.9853 | $-1.2 \times 10^{-3}$ | $5.3 \times 10^{-5}$ | 0.9704 | $4.2 \times 10^{-5}$ | OOM                  | OOM    | OOM                  | OOM                  | OOM    | OOM                  | OOM                  | OOM    | OOM                  | OOM    | OOM    | OOM                  | -95.7%  | +59.6% | -124.3% |
| IACS            | 0.0013               | 0.9764 | -0.0062              | $6.3 \times 10^{-5}$ | 0.9114 | $4.8 \times 10^{-4}$ | $2.8 \times 10^{-5}$ | 0.9963 | $-1.4 \times 10^{-3}$ | $1.6 \times 10^{-4}$ | 0.7934 | $1.2 \times 10^{-4}$ | $5.3 \times 10^{-8}$ | 0.9394 | $1.7 \times 10^{-7}$ | $2.3 \times 10^{-8}$ | 0.9943 | $4.3 \times 10^{-8}$ | $2.8 \times 10^{-7}$ | 0.9474 | $1.9 \times 10^{-7}$ | OOM    | OOM    | OOM                  | -93.5%  | +48.4% | -107.2% |
| CS-TGN          | $8.3 \times 10^{-4}$ | 0.9534 | $3.9 \times 10^{-4}$ | $1.7 \times 10^{-5}$ | 0.8043 | $4.3 \times 10^{-4}$ | OOM                  | OOM    | OOM                   | OOM                  | OOM    | OOM                  | OOM                  | OOM    | OOM                  | OOM                  | OOM    | OOM                  | OOM                  | OOM    | OOM                  | OOM    | OOM    | OOM                  | -88.7%  | +50.1% | -97.6%  |
| UTCS            | $8.3 \times 10^{-4}$ | 0.7904 | 0.0042               | $7.3 \times 10^{-4}$ | 0.7654 | 0.0093               | $1.4 \times 10^{-4}$ | 0.7534 | 0.0012                | OOM                  | OOM    | OOM                  | OOM                  | OOM    | OOM                  | OOM                  | OOM    | OOM                  | OOM                  | OOM    | OOM                  | OOM    | OOM    | OOM                  | -63.0%  | +29.7% | -65.4%  |

\* A community is considered to be of higher quality when it exhibits a **higher TD**, a **lower TC**, and a **higher TM**.

the lightweight regression models at each node to better capture subtle temporal dynamics and structural properties within specific sub-intervals, thereby refining the coreness estimation.

**Figure 27: Effect of  $p$  on prediction accuracy**

**Exp-19: Effect of  $x$  on Prediction Accuracy of HM-Index.** The effect of the leaf node threshold  $x$ , used when constructing the HM-Index, on prediction accuracy is shown in Figure 28. As shown in the figure, prediction accuracy slightly declines as  $x$  increases from 20 to 60 across all datasets. A smaller  $x$  yields finer temporal partitions in the HM-Index leaf nodes, allowing regression models to better capture short-term dynamics and structural variations in vertex coreness. In contrast, larger  $x$  leads to coarser intervals that may miss subtle temporal patterns, causing a minor accuracy drop. Nevertheless, accuracy remains above 90% for all  $x$  values.

**Figure 28: Effect of  $x$  on prediction accuracy**

**Exp-20: Lower-Tail Quality under Adverse Queries.** We examine the lower-tail behavior of all methods using the same 100 queries per dataset as Exp-1. For each dataset, we identify the five queries on which COCO performs worst, corresponding to the worst 5%, and report the mean results of all methods and metrics on this fixed query set.

Table 6 compares all methods under this lower-tail setting. COCO consistently achieves higher TD, lower TC, and higher TM than the baselines. Compared with QTCS, the average differences are 54.8% in TD, 30.8% in TC, and 60.1% in TM, exceeding the corresponding differences of 43.22%, 20.59%, and 48.58% in Table 3. Compared with the average COCO results, TD decreases by 27.78%–43.64%, TM decreases by 42.86%–61.82%. These results show that COCO maintains a larger advantage over the baselines on adverse queries.

These lower-tail results do not establish a per-query quality guarantee. Potential failure cases include query vertices near weak or overlapping community boundaries and query intervals with sparse or irregular interactions. In these settings, small errors in predicted coreness or temporal representations can alter the ranking of candidates with similar scores, which may change the returned community. The output should therefore be interpreted cautiously when the input provides little structural or temporal community signal. Exact methods suit applications requiring hard combinatorial guarantees, whereas learning-based methods such as COCO better suit large temporal graphs where exact per-query computation is impractical and empirical community quality is prioritized.